

LEARNING REUSABLE REASONING CRITICS FROM EXPERT-REFERENCED SUPERVISION

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ABSTRACT

Process-level reward models can score intermediate reasoning, but they typically require step-level annotations, preference data, or external verifiers. We study whether paired expert demonstrations can instead train a reusable reasoning critic without any of these. A naive expert-versus-policy classifier risks learning source or formatting cues instead of reasoning quality. We introduce the **Prefix-level Expert-Referenced Critic (PERC)**, which avoids this by using each expert trace three ways: as a positive example, as the reference answer for weakly labelling policy rollouts by agreement, and as the basis for targeted corruptions used as hard negatives, so that policy traces appear on both sides of the classifier and provenance alone cannot separate positives from negatives. Across GSM8K, MMLU-PRO, and MEDREASON, this critic captures a reusable reasoning-quality signal. As an **inference-time ranker**, it improves Best-of-16 pass@1 in all main reranking settings, transfers positively across all 27 source–target pairs with gains up to 12.7 points, complements consistency-based voting, and continues to improve pass@1 even for policies already trained with it. As a **process-level evaluator**, prefix-value changes localise human-labelled MEDREASON error units (two doctors, three final-year medical students) better than likelihood-based baselines and identify controlled GSM8K perturbations with up to 78.9% Hit@1. It is also a viable **training signal** for GRPO and remains competitive with imitation baselines without task-specific reward tuning. Overall, PERC shows that weak expert-referenced supervision yields reusable critics for inference-time selection and error diagnosis, while remaining feasible as a policy-training reward.¹

1 INTRODUCTION

Reasoning post-training for large language models (LLMs) often relies on reinforcement learning (RL) with external outcome verifiers or step-level supervision (Guo et al., 2025; Uesato et al., 2022; Lightman et al., 2023). In many practical settings, separate verifiers and process labels are difficult to design or expensive to collect, whereas expert demonstrations (from humans or powerful LLMs) are comparatively easy to obtain. This makes supervised fine-tuning (SFT) on expert traces attractive (Guo et al., 2025). However, SFT trains models to imitate observed reasoning traces rather than to evaluate policy-generated alternatives. As a result, when generation deviates from demonstrated paths, the model lacks an explicit critic to rank candidate traces or diagnose where reasoning fails (Setlur et al., 2025).

A natural alternative is to learn such a critic from expert demonstrations. A naive expert-versus-policy classifier, however, can learn source or formatting cues rather than reasoning quality. We therefore ask whether paired expert traces can be used as references for training a reusable prefix-level critic: policy rollouts are labelled by final-answer agreement with the paired expert answer, and targeted corrupted expert traces provide hard negatives that preserve expert style while altering reasoning-relevant content.

This raises a key question: can referencing expert answers and expert content, not just using expert traces directly as positives, train a critic that scores reasoning quality rather than source or formatting cues? Concretely, this means using agreement labelling and targeted corruption on top of direct

¹We provide the code for our experiments at https://anonymous.4open.science/r/expert_reasoning-C1E8/

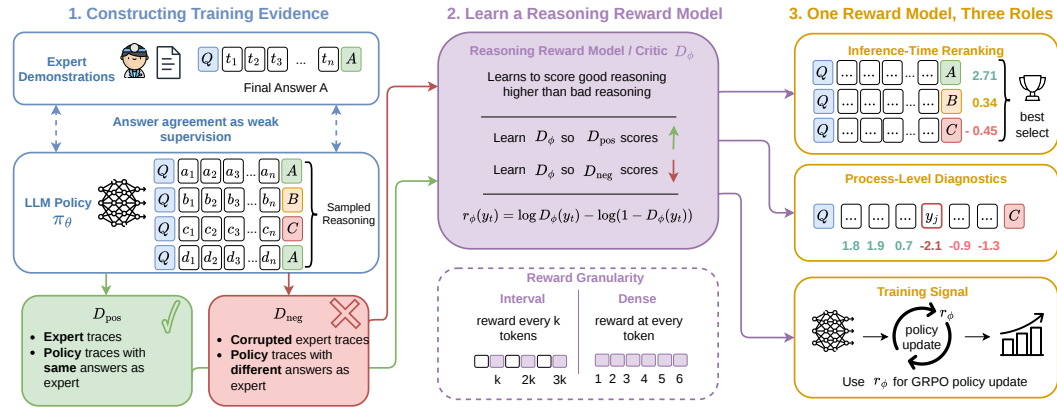


Figure 1: A demonstration-learned critic reused across three roles. PERC learns a prefix-level reasoning critic from expert-referenced trace evidence, without a separate external verifier or step-level annotations, and we evaluate the same frozen critic for inference-time reranking, error localisation, and policy optimisation.

positives. We evaluate such a critic in three roles, beginning with uses that do not require further policy updates:



We evaluate three roles for a learned reasoning critic:

1. **(D1) Inference-time ranking.** Selecting better sampled traces under a fixed test-time budget.
2. **(D2) Process-level feedback.** Localising where reasoning first becomes unreliable or incorrect.
3. **(D3) Policy-optimisation feasibility.** Testing whether the same diagnostic critic can double as an aggregate training reward without collapsing policy updates.

We introduce the Prefix-level Expert-Referenced Critic (PERC). PERC is not supervision-free: policy traces are labelled by agreement with the paired expert final answer. The setting we target is instead weaker than process supervision: no step-level annotations, no preference pairs, and no separate external verifier beyond the paired expert answer. PERC trains a prefix-level critic once from expert demonstrations, answer-agreement-labelled policy rollouts, and targeted corrupted expert traces. The critic is then frozen and reused for inference-time reranking ((D1)), process-level error localisation ((D2)), and policy optimisation ((D3)).

Contributions. We make three contributions:

1. **Expert-referenced prefix critic learning.** We train prefix-level reasoning critics from expert traces, answer-agreement-labelled policy rollouts, and targeted hard negatives at multiple granularities.
2. **One frozen critic, multiple roles.** We show that the same trained critic can be used for inference-time reranking, prefix-change error localisation, and aggregate policy optimisation.
3. **Empirical validation of critic reuse.** Across GSM8K, MEDREASON, and MMLU-PRO, we demonstrate that PERC delivers substantial gains in test-time reranking and fine-grained error diagnosis (both on controlled perturbations and expert clinician labels), while confirming its feasibility as an aggregate policy-training signal.

This makes the central contribution a reusable, process-level critic learned from expert-referenced trace evidence, rather than recovering an identifiable latent reward from expert behaviour alone. Figure 1 summarises the full pipeline.

2 PROBLEM FORMALISM

For a prompt $x \sim \mathcal{Q}$, we model an output as a reasoning trajectory $y = (y_1, \dots, y_T)$ with autoregressive likelihood $p_\theta(y \mid x) = \prod_{t=1}^T \pi_\theta(y_t \mid x, y_{<t})$. We write the prefix after token t as $h_t = (x, y_{\leq t})$.

Prefix value and process score. The true step-level reward is unobserved, so PERC learns a prefix critic $D_\phi(h_t) \in (0, 1)$ for whether prefix h_t belongs to the positive expert-referenced trace class. We convert this probability into a log-odds prefix value $V_\phi(h_t) = \log D_\phi(h_t) - \log(1 - D_\phi(h_t))$ and aggregate supervised positions as $R_\phi(y; m) = Z_m^{-1} \sum_{t=1}^T m_t V_\phi(h_t)$, where $m_t \in \{0, 1\}$ and $Z_m = \sum_t m_t$. Dense, interval, and sparse rewards differ only in the mask m .

Expert-referenced critic. PERC does not recover an identifiable expert reward and does not use online adversarial critic updates. Instead, we first train a prefix critic on constructed positive and negative trace sets $\mathcal{D}_+$ and $\mathcal{D}_-$, defined in Section 4. After training, the critic is frozen and reused:

$$\hat{\phi} = \arg \min_{\phi} \mathcal{L}_D(\phi; \mathcal{D}_+, \mathcal{D}_-), \quad \max_{\theta} \mathbb{E}_{y \sim \pi_\theta} [R_{\hat{\phi}}(y; m)]. \quad (1)$$

Thus, the learned score is a classifier-derived prefix-value signal under expert-referenced supervision, not a ground-truth reward recovered through adversarial inverse reinforcement learning.

3 RELATED WORK

Outcome and process supervision. Outcome-supervised reasoning RL optimises a terminal reward,

$$\max_{\theta} \mathbb{E}_{y \sim \pi_\theta} [R(y; m^{\text{out}})], \quad m_t^{\text{out}} = \mathbb{I}[t = T]. \quad (2)$$

where R is usually derived from an external or programmatic outcome verifier (Guo et al., 2025; Uesato et al., 2022; Yu et al., 2025; YuYue et al., 2025). Process reward models use denser masks and can improve reasoning supervision (Uesato et al., 2022; Lightman et al., 2023), with search variants further improving exploration (Zelikman et al., 2022; Yuan et al., 2023; Singh et al., 2024; Hosseini et al., 2024; Silver et al., 2017). PRIME and IPVRM reduce the need for step labels by deriving dense or prefix-level signals from outcome-labelled rollouts (Cui et al., 2025; Gao et al., 2026). PERC is closest to this prefix/process-critic line, but constructs its critic data relative to paired expert traces: expert demonstrations, answer-agreement-labelled policy rollouts, and targeted corrupted expert traces.

Preference-based reward learning. Preference-based RL learns a reward model R_ϕ , typically outcome-level, from pairwise preferences between a winning response y^w and a losing response y^l for the same prompt x :

$$\max_{\phi} \mathbb{E} [\log \sigma(R_\phi(y^w; m^{\text{out}}) - R_\phi(y^l; m^{\text{out}}))] , \quad \max_{\theta} \mathbb{E}_{y \sim \pi_\theta} [R_\phi(y; m^{\text{out}})].$$

These methods use preference data rather than verifier-labelled rewards (Christiano et al., 2023; Rafailov et al., 2024). They can support reranking and policy optimisation, but are typically response-level and less direct for localisation.

Demonstration-based learning. Supervised fine-tuning on expert traces optimises

$$\max_{\theta} \mathbb{E}_{y^E \sim p_E} \left[\frac{1}{T} \sum_t \log \pi_\theta(y_t^E) \right].$$

This often improves reasoning quality (Hinton et al., 2015; Guo et al., 2025; Kang et al., 2023; Kujanpää et al., 2025; Xu et al., 2025), but learns no explicit critic for policy-generated trajectories and therefore does not natively support reranking, localisation, or reward-based policy optimisation (Setlur et al., 2025). It is also purely off-policy: the objective never uses samples from the model being trained. PERC trains a critic with policy rollouts as positive and negative evidence, then uses on-policy GRPO updates from the evolving policy. Continual self-distillation is complementary (Shenfeld et al., 2026). Broader demonstration-based reward-learning methods include Inverse-RLignment/AfD (Sun and Schaar, 2025) and PARED (Wiliński et al., 2026); PERC differs by learning a prefix-level reasoning critic and evaluating its reuse for ranking, localisation, and optimisation.

Source-label distillation. Adversarial distillation methods such as GAD distinguish teacher and student generations, often alongside an outcome verifier (Ye et al., 2025; Lee et al., 2025):

$$\max_{\phi} \min_{\theta} \mathbb{E}_{y^E \sim p_E} [R_{\phi}(y^E; m^{\text{out}})] - \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y; m^{\text{out}})], \quad \max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y; m^{\text{out}}) + R(y; m^{\text{out}})].$$

Such source-label critics can reward teacher provenance or style. PERC instead puts policy traces on both sides of the classifier through answer agreement, and places expert-style corruptions in the negative class. Its contribution is therefore not adversarial learning itself, but the expert-referenced construction of a reusable prefix critic. We evaluate whether this critic supports reranking ①1, localisation ①2, and policy optimisation ①3.

We summarise this comparison in Table 1.

Method Class	Example Works	Objective / Training Target	Supervision Data	Learned Signal	①1 Rank	①2 Localise	①3 Optimise
Verifier / outcome supervision							
Outcome-supervised RL	Guo et al. (2025) Yu et al. (2025) YuYue et al. (2025)	$\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}} [R(y; m^{\text{out}})]$	Outcome verifier	Terminal reward	✓	✗	✓
Outcome-labelled process critics	Cui et al. (2025) Gao et al. (2026)	$\max_{\phi} \mathbb{E}_{(h_t, z)} [z \log q_{\phi}(h_t) + (1 - z) \log(1 - q_{\phi}(h_t))]$ $r_t \approx q_{\phi}(h_t) - q_{\phi}(h_{t-1})$	Outcome-labelled rollouts	Prefix / process value	✓	✓	✓
Process-supervised RL	Uesato et al. (2022) Lightman et al. (2023)	$\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}} [R(y; m)]$	Process labels	Step-level reward	✓	✓	✓
Preference supervision							
Preference-supervised RL	Christiano et al. (2023) Rafailov et al. (2024)	$\max_{\phi} \mathbb{E}_{(y^w, y^l)} [\log \sigma(R_{\phi}(y^w; m^{\text{out}}) - R_{\phi}(y^l; m^{\text{out}}))]$ then $\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y; m^{\text{out}})]$	Preference pairs	Response reward	✓	✗	✓
Demonstration / source supervision							
SFT	Hinton et al. (2015) Guo et al. (2025) Xu et al. (2025)	$\max_{\theta} \mathbb{E}_{y^E \sim \mathcal{D}_E} [\frac{1}{T} \sum_t \log \pi_{\theta}(y_t^E)]$	Demos	Policy likelihood	✗	✗	✗
Demo reward learning	Sun and Schaar (2025) Wiliński et al. (2026)	$\max_{\phi} \mathbb{E}_{y^E \sim p_E} [\log D_{\phi}(y^E)]$ $+ \mathbb{E}_{y \sim \pi_{\theta}} [\log(1 - D_{\phi}(y))]$ then $\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y)]$	Demos + policy samples	Response-level demo reward	✓	✗	✓
Source-label distillation	Ye et al. (2025) Lee et al. (2025)	$\max_{\phi} \min_{\theta} (\mathbb{E}_{y^E \sim p_E} [R_{\phi}(y^E; m^{\text{out}})] - \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y; m^{\text{out}})])$ then $\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y; m^{\text{out}}) + R(y; m^{\text{out}})]$	Teacher / student source	Source / provenance reward	(✓)	✗	✓
Expert-referenced supervision							
PERC	this work	$\hat{\phi} = \arg \min_{\phi} \mathcal{L}_D(\phi; \mathcal{D}_+, \mathcal{D}_-)$ then reuse $R_{\hat{\phi}}(y; m)$	Demos + answer agreement + corruptions	Expert-ref. prefix critic	✓	✓	✓

Table 1: **Related work comparison.** Methods are grouped by supervision regime while preserving the ①1–①3 roles used in our experiments. The learned-signal column distinguishes terminal rewards, supervised step rewards, outcome-labelled prefix/process critics, response-level demonstration rewards, source/provenance rewards, and PERC’s expert-referenced prefix critic. Checkmarks indicate studied uses, not identical assumptions; parenthesised checkmarks indicate partial support.

4 METHOD

Our goal is to learn a reusable process-level reasoning signal from expert-referenced trajectories, without a separate external verifier, step-level labels, or preference pairs. PERC trains a prefix critic once on constructed positive and negative trace evidence, then reuses its induced prefix values to rank fixed samples, localise error spans, and, after aggregation, update a policy. The critic is therefore not an identifiable expert reward, but a classifier-derived expert-referenced signal.

4.1 TRAINING EVIDENCE CONSTRUCTION

A naive split (all expert positive, all policy negative) encourages style rather than reasoning-quality discrimination (Ye et al., 2025; Lee et al., 2025). We instead use answer agreement with the paired expert trace as weak outcome supervision. Policy rollouts for critic training are sampled from the initial policy π_0 . For prompt x , let y^E be the paired expert trace and let $\mathcal{O}(y)$ be the normalised final-answer string extracted from y ; agreement is exact string equality after normalisation, with extraction details in Appendix B:

$$\mathcal{D}_{\text{pos}} = \{y \sim p_E\} \cup \{y \sim \pi_0 \mid \mathcal{O}(y) = \mathcal{O}(y^E)\}.$$

Including answer-agreeing policy rollouts in $\mathcal{D}_{\text{pos}}$ puts policy text on both sides of the classifier, reducing source and formatting shortcuts. For negatives, we use answer-mismatched policy rollouts and paired counterfactual hard negatives. Given an expert trace y^E , the corruption operator $\mathcal{C}(y^E)$ preserves the prompt, formatting, source, and most content while minimally altering a reasoning-relevant property, such as a local arithmetic step, numerical value, or answer-bearing intermediate conclusion (Appendix B):

$$\mathcal{D}_{\text{neg}} = \{y \sim \pi_0 \mid \mathcal{O}(y) \neq \mathcal{O}(y^E)\} \cup \{\mathcal{C}(y^E) \mid y^E \in \mathcal{D}^E\}.$$

We assign the resulting labels at the trace level and apply them to the supervised token positions selected by m_t . Thus, the learned prefix values are induced process signals from weak expert-referenced labels, not observed step-level correctness labels. Critic batches are class-balanced; corruptions and clean expert traces are generated at a 1:1 rate (Appendix B).

4.2 REASONING CRITIC AND REWARD DEFINITION

We train a prefix classifier $D_\phi(h_t) \in (0, 1)$, where $h_t = (x, y_{\leq t})$, to distinguish prefixes from $\mathcal{D}_{\text{pos}}$ and $\mathcal{D}_{\text{neg}}$. Because supervision is trace-level, $D_\phi(h_t)$ should not be read as a direct label for whether token y_t is correct. Instead, it estimates how strongly the partial trace resembles the positive expert-referenced class. A binary mask $m_t \in \{0, 1\}$ specifies where this supervision is applied. We evaluate two process-level granularities in the main experiments: *Interval* ($m_t = 1$ every k tokens) and *Fully Dense* ($m_t = 1$ for all t). The same formulation also admits a *Sparse Outcome* variant ($m_t = \mathbb{I}[t = T]$), which scores only the final position. Coarser masks are typically more stable for optimisation, while denser masks provide finer process feedback for localisation.

The critic is trained by minimising the length-normalised binary cross-entropy:

$$\mathcal{L}_D(\phi) = -\mathbb{E}_{y \sim \mathcal{D}_{\text{pos}}} \left[Z_m^{-1} \sum_{t=1}^T m_t \log D_\phi(h_t) \right] - \mathbb{E}_{y \sim \mathcal{D}_{\text{neg}}} \left[Z_m^{-1} \sum_{t=1}^T m_t \log (1 - D_\phi(h_t)) \right]. \quad (3)$$

After fitting the critic, we convert probabilities into prefix values:

$$V_\phi(h_t) = \log D_\phi(h_t) - \log(1 - D_\phi(h_t)). \quad (4)$$

For localisation, the local process signal is the value change $\delta_\phi(t) = V_\phi(h_t) - V_\phi(h_{t-1})$; large negative changes indicate prefixes where the trace becomes less consistent with the positive class.

4.3 POLICY OPTIMISATION WITH LEARNED REWARDS

For policy optimisation, we keep the trained critic D_ϕ fixed and optimise π_θ with Group Relative Policy Optimisation (GRPO) (Shao et al., 2024). At each policy iteration, we freeze θ_{old} , sample a group of G traces $\{y^{(g)}\}_{g=1}^G \sim \pi_{\theta_{\text{old}}}$ for prompt x , and update θ while keeping θ_{old} fixed.

The prefix value $V_\phi(h_t)$ is directly available only at supervised positions. For *interval* masks, each supervised value is backfilled to the preceding unsupervised tokens; the same rule would apply to the sparse outcome variant:

$$t' = \min\{k \geq t \mid m_k = 1\}, \quad r_\phi(y_t) = V_\phi(h_{t'}).$$

For *dense*, $r_\phi(y_t) = V_\phi(h_t)$.

For policy optimisation, we aggregate prefix values into a length-normalised sequence score following Cetin et al. (2025) and Cui et al. (2025):

$$\bar{r}^{(g)} = \frac{1}{T_g} \sum_{t=1}^{T_g} r_\phi(y_t^{(g)}), \quad A^{(g)} = \frac{\bar{r}^{(g)} - \mu}{\sigma + \epsilon}. \quad (5)$$

We assign the same sequence-level advantage $A^{(g)}$ to all tokens in generation g and optimise the PPO clip loss (Schulman et al., 2017) on policy-generated traces. Thus, ⑬ tests the critic as an aggregate policy-training signal, while ⑫ evaluates its token-level structure through prefix-value changes.

Appendix A provides implementation details on critic training, batching, reward clipping, and policy optimisation; Algorithm 1 summarises the full procedure.

5 EXPERIMENTS

We evaluate PERC on GSM8K (Cobbe et al., 2021), MEDREASON (Wu et al., 2025), and MMLU-PRO (Wang et al., 2024). These benchmarks stress complementary regimes: multi-step arithmetic with human-written demonstrations, medical-domain multiple-choice transfer with plausible distractors, and broad scientific reasoning. Our base policies are open-weight instruction-tuned variants not trained for reasoning: Qwen2.5 (7B) (Bai et al., 2023), Llama3 (8B) (Touvron et al., 2023), and Qwen3-4B (Yang et al., 2025). We instantiate each policy as a text generator and the reasoning critic as a prefix/token classifier.

Because PERC’s central output is a reusable frozen critic, we first evaluate critic reuse before policy optimisation. Section 5.1 tests inference-time ranking on fixed candidate sets and transfer across tasks/backbones. Section 5.2 tests whether prefix-value changes localise medical-expert-labelled natural errors and controlled perturbations. Section 5.3 then asks whether aggregating the same critic also provides a competitive policy-optimisation signal. Appendix B contains implementation details.

For each backbone and dataset, we train PERC critics with *interval* ($k = 15$) and *dense* masks using the evidence construction in Section 4, then keep the critic fixed for all downstream uses. The experiments below then evaluate the resulting critic in three ways. We begin with inference-time use, where the trained critic is held fixed and used only to score candidate traces. We then test process-level localisation via prefix-value changes, and finally report policy-optimisation results using the same critics as aggregate rewards.

5.1 INFERENCE-TIME CRITIC-GUIDED RERANKING

RQ1: *Can the learned critic select better traces from a fixed candidate set at inference time?*

Experimental setup. Given a trained PERC critic, we rank N sampled traces by their prefix-value score and return the highest-scoring one. Random selection from the same candidate set equals the generator’s expected pass1, so gains isolate selection quality without further policy updates.

Fixed-candidate ranking and transfer. Table 2 uses fixed Qwen2.5-7B SFT candidates, so only the scoring critic changes. Diagonal entries test same-dataset ranking, off-diagonal entries test transfer, and the dashed divider separates same-backbone from out-of-backbone critics. All 27 source–target pairs improve, with gains up to +12.7 pp.

Policy Dataset	Baseline	Qwen2.5-7B			Llama3.1-8B			Qwen3-4B		
		GSM8K	MMLU-PRO	MEDREASON	GSM8K	MMLU-PRO	MEDREASON	GSM8K	MMLU-PRO	MEDREASON
GSM8K	71.1 $\pm$ 1.7	79.9 ($\uparrow$ +8.8)	79.9 ($\uparrow$ +8.8)	75.1 ($\uparrow$ +4.0)	82.5 ($\uparrow$ +11.4)	78.8 ($\uparrow$ +7.7)	81.2 ($\uparrow$ +10.1)	83.8 ($\uparrow$ +12.7)	83.3 ($\uparrow$ +12.2)	78.9 ($\uparrow$ +7.8)
MMLU-PRO	48.1 $\pm$ 1.9	49.0 ($\uparrow$ +0.9)	51.3 ($\uparrow$ +3.2)	51.8 ($\uparrow$ +3.7)	53.8 ($\uparrow$ +5.7)	53.8 ($\uparrow$ +5.7)	48.9 ($\uparrow$ +0.8)	56.8 ($\uparrow$ +8.7)	55.3 ($\uparrow$ +7.2)	54.3 ($\uparrow$ +6.1)
MEDREASON	66.3 $\pm$ 2.1	67.1 ($\uparrow$ +0.8)	69.5 ($\uparrow$ +3.2)	68.5 ($\uparrow$ +2.2)	70.1 ($\uparrow$ +3.8)	71.9 ($\uparrow$ +5.6)	78.0 ($\uparrow$ +11.7)	71.4 ($\uparrow$ +5.1)	69.9 ($\uparrow$ +3.6)	71.9 ($\uparrow$ +5.6)

Table 2: **Best-of-16 cross-critic transfer reranking values (%)**. All candidates come from the same frozen Qwen2.5-7B SFT generator; only the scoring critic varies, isolating critic quality from generator changes. Cells report PERC-guided pass@1, with gains over random selection in parentheses. Italics mark same-dataset critics; the dashed divider separates same-backbone from out-of-backbone critics. Table 3 instead varies the generator by row.

Reranking behaviour. Figure 2 shows both when and why critic-guided selection helps. Across $N \in \{2, 3, 5, 8, 16\}$, critic reranking is strongest at low budgets, while reward-weighted majority voting is best at larger N . The critic also separates correct from incorrect traces ($t = 87.43, p < 0.001$), explaining the gains in Table 3; full budget and separation results are in Appendix Tables 12–16 and Appendix Figures 6–14.

In-distribution reranking. Table 3 differs from Table 2: candidates are sampled from each GRPO-optimised policy, not from a shared frozen SFT pool. Thus, the Random column is that policy’s own pass@1 and varies by backbone, dataset, and granularity. PERC-guided reranking still improves all settings at $N = 16$, with gains up to +8.7 pp (Qwen3-4B+*interval*, MMLU-PRO) and +6.4 pp (Llama3.1-8B+*dense*, GSM8K); all MEDREASON gains are positive (+0.7 to +5.8 pp). This shows that the critic remains useful for selection even after optimising the policy.

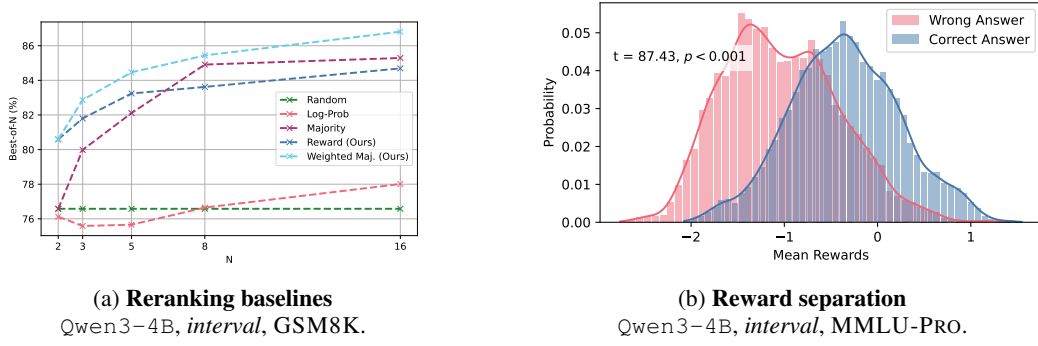


Figure 2: **Inference-time behaviour of learned critics.** 2a: low-budget gains from reward-guided reranking; at higher N , reward-weighted majority voting is strongest. 2b: correct traces receive higher critic scores than incorrect traces ($t = 87.43, p < 0.001$).

Method	Granularity	GSM8K			MMLU-PRO			MEDREASON			Avg. Δ
		Random	PERC	Δ	Random	PERC	Δ	Random	PERC	Δ	
Qwen2.5-7B	Interval	90.2 $\pm$ 1.3	92.2 $\pm$ 1.4	($\uparrow$ +2.0)	50.6 $\pm$ 1.9	53.1 $\pm$ 2.5	($\uparrow$ +2.5)	67.2 $\pm$ 2.5	68.1 $\pm$ 2.9	($\uparrow$ +0.9)	+1.8
	Dense	89.7 $\pm$ 1.3	92.2 $\pm$ 1.4	($\uparrow$ +2.5)	43.8 $\pm$ 2.0	46.0 $\pm$ 2.5	($\uparrow$ +2.2)	68.7 $\pm$ 2.4	70.2 $\pm$ 2.8	($\uparrow$ +1.5)	+2.1
Llama3.1-8B	Interval	77.9 $\pm$ 1.7	82.2 $\pm$ 2.0	($\uparrow$ +4.3)	36.6 $\pm$ 1.8	38.1 $\pm$ 2.4	($\uparrow$ +1.6)	76.3 $\pm$ 2.1	81.4 $\pm$ 2.5	($\uparrow$ +5.1)	+3.7
	Dense	65.7 $\pm$ 2.0	72.1 $\pm$ 2.4	($\uparrow$ +6.4)	37.9 $\pm$ 1.8	39.7 $\pm$ 2.5	($\uparrow$ +1.9)	75.1 $\pm$ 2.1	80.9 $\pm$ 2.4	($\uparrow$ +5.8)	+4.7
Qwen3-4B	Interval	90.0 $\pm$ 1.4	92.9 $\pm$ 1.4	($\uparrow$ +2.9)	53.5 $\pm$ 2.4	62.2 $\pm$ 2.4	($\uparrow$ +8.7)	69.1 $\pm$ 2.6	71.4 $\pm$ 2.9	($\uparrow$ +2.3)	+4.6
	Dense	92.6 $\pm$ 1.2	94.0 $\pm$ 1.3	($\uparrow$ +1.5)	55.1 $\pm$ 2.3	59.2 $\pm$ 2.4	($\uparrow$ +4.1)	69.2 $\pm$ 2.5	69.9 $\pm$ 2.8	($\uparrow$ +0.7)	+2.1

Table 3: **Best-of-16 in-distribution reranking performance (%)**. Candidates come from the GRPO-optimised policies in Section 5.3, unlike the fixed SFT pool in Table 2. PERC is compared with random selection from the same samples; Δ is the percentage-point gain, and Avg. Δ averages across datasets.



Takeaway. RQ1 is answered positively: the learned critic is a reusable inference-time ranking signal that improves all 27 transfer settings and complements consistency-based voting under fixed budgets.

5.2 REASONING ERROR LOCALISATION

RQ2: *Do prefix-value changes from the learned critic localise where reasoning first fails?*

Reranking shows that the learned critic scores complete traces. We next ask whether its prefix structure is locally meaningful: when reasoning first becomes invalid, does the critic value drop near the error?

Experimental setup. We use two complementary localisation tests. For natural errors, three final-year medical students and two doctors annotate each of 10-25 incorrect Qwen2.5-7B SFT generations from MEDREASON, marking the first reasoning unit that is medically incorrect, unsupported, contradictory, or misleading. We exclude non-localisable cases; if the reasoning is clinically correct but the final option is wrong, the target is the final unit. More information can be found in Appendix A.4.

We also perform a second controlled experiment for mathematical reasoning on GSM8K; we start from expert traces and insert one targeted arithmetic perturbation while preserving the prompt, formatting, and most surrounding text: an operator flip, numeric-literal offset, or replacement by an earlier intermediate value (Appendix B). This gives the exact injected error token, testing whether value changes localise the intervention rather than only detecting a wrong final answer.

Hit metrics. For each signal a , we score token transitions by $d_t^{(a)} = \max(0, z_{t-1}^{(a)} - z_t^{(a)})$, where $z_t^{(a)}$ is the PERC prefix value, token log-probability, or negative entropy at prefix h_t . The highest-scoring transition is the predicted error. For medical-expert labels, we map this token prediction to its numbered reasoning unit: Hit@1 requires the medical-expert-marked first-error unit, while Hit@ ± 1

also accepts one adjacent unit. For controlled GSM8K, Hit@ k requires the predicted transition to fall within k tokens of the injected perturbation; thus Hit@1/Hit@7 use $\pm 1/\pm 7$ token windows.

Setting	Metric	Chance	Log-prob.	Entropy	Interval (ours)	Dense (ours)
Human-labelled MEDREASON	Hit@1 (unit)	8.1	4.5	9.1	30.0	<u>16.4</u>
	Hit@ ± 1 (unit)	20.0	17.3	21.8	40.9	<u>28.2</u>
Controlled GSM8K	Hit@1 (token)	11.9	25.9	0.2	78.9	<u>62.0</u>
	Hit@7 (token)	32.0	26.9	1.9	78.9	<u>70.3</u>

Table 4: **Error localisation.** Hit rates (%). Medical-expert MEDREASON is unit-level ($n=110$); controlled GSM8K reports the Qwen3-4B synthetic perturbation slice ($n=861$).

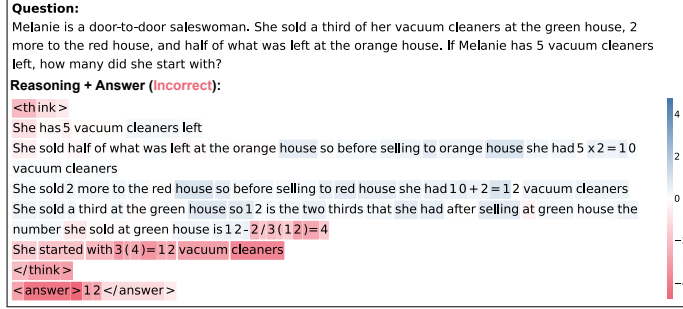


Figure 3: **Qualitative example.** The dense PERC prefix value drops at the first erroneous step.

Localisation results. Table 4 shows that PERC prefix-value changes localise medical-expert-marked errors better than likelihood-based baselines. Interval rewards are best for exact-unit hits (30.0% vs. 8.1% random) and within one neighbouring unit (40.9% vs. 20.0%). The controlled GSM8K diagnostic isolates the first injected reasoning edit; for Qwen3-4B, interval reaches 78.9% Hit@1/Hit@7 versus 25.9%/26.9% for log-probability. Figure 3 shows the prefix value staying high on valid steps and dropping at the first error; Appendix Figure 15 gives further correct/incorrect reward divergence examples.

Takeaway. RQ2 is answered positively: learned prefix-value changes improve natural-error localisation over token log-probabilities and entropy.

5.3 USING THE SAME CRITIC FOR POLICY OPTIMISATION

RQ3: *Can the diagnostic prefix critic also serve as a viable policy-optimisation signal?*

Experimental setup. Finally, we test whether the same frozen critic can also support policy updates. For each backbone, we optimise π_θ with *interval* ($k = 15$) and *dense* PERC scores aggregated into a sequence-level GRPO advantage (Section 4.3). This tests aggregate reward quality, not token-specific credit assignment.

Method	Granularity	GSM8K			MMLU-PRO			MEDREASON		
		Qwen2.5-7B	Llama3.1-8B	Qwen3-4B	Qwen2.5-7B	Llama3.1-8B	Qwen3-4B	Qwen2.5-7B	Llama3.1-8B	Qwen3-4B
SFT	—	71.1 $\pm$ 1.6	67.5 $\pm$ 2.1	77.5 $\pm$ 1.6	48.1 $\pm$ 1.9	47.2 $\pm$ 1.9	53.9 $\pm$ 2.0	66.3 $\pm$ 2.2	79.5 $\pm$ 2.0	67.4 $\pm$ 2.3
PERC	Interval	90.2 $\pm$ 1.3	77.9 $\pm$ 1.7	<u>90.0</u> $\pm$ 1.4	50.6 $\pm$ 1.9	36.6 $\pm$ 1.8	53.5 $\pm$ 2.4	67.2 $\pm$ 2.5	<u>76.3</u> $\pm$ 2.1	<u>69.1</u> $\pm$ 2.6
	Dense	<u>89.7</u> $\pm$ 1.3	65.7 $\pm$ 2.0	92.6 $\pm$ 1.2	43.8 $\pm$ 2.0	<u>37.9</u> $\pm$ 1.8	55.1 $\pm$ 2.3	68.7 $\pm$ 2.4	75.1 $\pm$ 2.1	69.2 $\pm$ 2.5

Table 5: **Held-out pass@1 in the no-external-verifier setting.** Bold/underlined mark best/second-best across SFT and PERC variants shown in the main text. Values are the mean $\pm$ half-width of the bootstrapped 95% confidence interval.

Policy-optimisation performance. Table 5 shows that PERC provides a competitive aggregate reward signal, but is not uniformly better than SFT. Selecting the best granularity per backbone-dataset pair, PERC improves over SFT in 7/9 settings. Gains are strongest on GSM8K (3/3 backbones) and

positive for 2/3 backbones on both MMLU-PRO and MEDREASON; the main underperforming cases are Llama3.1-8B on MMLU-PRO and MEDREASON. We interpret this as a feasibility result: the learned critic can be aggregated into a useful policy-training reward, while its main advantage is that the same critic also supports inference-time reranking and process-level localisation.

Reranking after policy optimisation. The policies in Table 5 also generate the candidates used in Table 3. Thus, Table 3 tests whether the frozen critic still helps after its aggregate signal has already updated the policy. It does: PERC-guided reranking improves pass@1 in all settings, by up to +8.7 pp (Qwen3-4B+*interval*, MMLU-PRO), showing that the critic remains a complementary inference-time signal rather than being fully absorbed into the policy weights.

Additional baselines. Table 6 compares PERC with direct answer-agreement GRPO and Generative Adversarial Distillation (GAD) (Ye et al., 2025) on Qwen2.5-7B. Answer-agreement GRPO provides a dedicated binary terminal reward at step T . By design, it optimises exclusively for terminal pass@1 on tasks with programmatic string extraction, but yields no reusable prefix artefact for ① reranking or ② localisation. In contrast, PERC is designed first and foremost as a reusable process critic: it approaches or matches the performance of terminal answer-agreement GRPO in training, while uniquely unlocking fine-grained step diagnosis and test-time reranking that terminal RL cannot provide. These signals are therefore complementary rather than mutually exclusive: SFT, answer-agreement GRPO, and PERC can be combined when exact answer matching is available, while PERC provides the reusable critic studied in RQ1 and RQ2. Compared with the GAD-style source-label baseline, which uses teacher/student trace provenance as its signal, PERC gives stronger pass@1 and larger reranking gains.

For completeness, Appendix C presents ablations on expert corruptions, group size, clipping, and policy-vs-expert-only separation.

Takeaway. RQ3 shows that the critic can serve as a useful aggregate policy-optimisation reward. Its main value, however, is broader: the same critic remains reusable for inference-time selection and process-level diagnosis, rather than being absorbed into the trained policy.

Method	Granularity	GSM8K	MMLU-PRO	MEDREASON	① Avg. Δ	② Localise
SFT	–	71.1 $\pm$ 1.6	48.1 $\pm$ 1.9	66.3 $\pm$ 2.2	(X)	(X)
Ans-w-agr. GRPO	Terminal	90.0 $\pm$ 1.4	53.5 $\pm$ 2.0	69.2 $\pm$ 2.4	(X)	(X)
Source-based	Terminal	76.2 $\pm$ 1.4	49.3 $\pm$ 2.1	67.8 $\pm$ 2.4	✓ (↑ +0.3)	(X)
PERC	Interval	90.2 $\pm$ 1.3	50.6 $\pm$ 1.9	67.2 $\pm$ 2.5	✓ (↑ +1.8)	✓
PERC	Dense	89.7 $\pm$ 1.3	43.8 $\pm$ 2.0	68.7 $\pm$ 2.4	✓ (↑ +2.1)	✓

Table 6: **Additional baselines.** Pass@1 is reported for SFT, answer-agreement GRPO, GAD, and PERC.

6 LIMITATIONS

PERC has several limitations. First, PERC is more expensive than SFT because it trains a separate prefix critic and then uses it for policy optimisation; Appendix Table 7 reports 3–4 $\times$ higher cost in representative runs. Second, answer agreement is weak supervision: it can miss flawed reasoning with a correct final answer and penalise partially correct traces. It is also easiest to apply on benchmarks with extractable normalised answers; open-ended settings would require more robust answer extraction or semantic matching. Third, denser rewards provide finer process feedback but are less stable, with occasional reward–correctness decoupling. Finally, our GRPO update aggregates prefix scores into one sequence-level advantage rather than using token-specific returns; improving this interface is future work.

7 CONCLUSION

PERC learns reusable reasoning critics from expert-referenced traces without a separate external verifier or step-level annotations. Across three datasets, the critic improves Best-of-16 reranking in all settings, transfers across all 27 source–target pairs, localises medical-expert-marked MEDREASON errors and controlled GSM8K perturbations better than likelihood baselines, and remains competitive when aggregated for policy optimisation. These results support weak expert-referenced supervision as a route to reusable critics for ranking, diagnosis, and training.

AI USE STATEMENT

In this work, we used generative AI tools to obtain demonstrations for solving reasoning tasks, implement and maintain the code for running experiments, suggest a structure for the paper, and prepare plots for the figures. We have not used generative AI tools to help develop theoretical models or conceptual frameworks, to propose or refine hypotheses, to design our research methodology, to support qualitative and thematic data analysis, or to interpret results, and the generation of synthetic datasets and assistance with translation are not applicable to this work. Additionally, we have used generative AI tools to edit individual paragraphs in the paper to improve readability. We reviewed all AI-assisted work: all AI-edited text was reviewed by authors to confirm it accurately reflects our intended meaning and claims. We verified the plotted values against matching plots from wandb. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

We provide complete definitions of PERC and its implementation. The appendix documents the model checkpoints, datasets, prompts, data splits, generation settings, optimisation hyperparameters, learning rates, gradient-clipping choices, training durations, and evaluation procedures used in each experiment. Unless otherwise stated, we report the 95% bootstrapped confidence interval score (1000 resamples). All matched comparisons use the same training and evaluation pipeline, differing only in the factors under investigation. The code can be found at https://anonymous.4open.science/r/expert_reasoning-C1E8/

ETHICS STATEMENT

Our study involves medical-expert annotation of model-generated reasoning traces. Annotators were asked to identify the first medically incorrect, unsupported, contradictory, or misleading reasoning unit in incorrect MEDREASON generations. They did not interact with deployed systems, patients, or private medical records; all annotated items were model outputs derived from benchmark questions. We collected only task annotations and did not report personally identifying information. We informed annotators about the purpose of the study, and participation was voluntary. The results are intended to evaluate reasoning-error localisation, not to validate the generated medical reasoning for clinical use.

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A ADDITIONAL METHOD DETAILS

A.1 TRAINING A PREFIX EXPERT-REFERENCED CRITIC

Several considerations arise when training this critic for LLMs. First, trajectories are discrete autoregressive token sequences, so prefix-level discriminator scores must be converted into stable rewards for policy-gradient updates. Second, a naive expert-vs-policy discriminator would mainly learn source or formatting cues; PERC therefore uses answer agreement and corrupted expert traces to construct weak positive and negative reasoning evidence. Third, we must explicitly control reward granularity: sparse, interval, and dense masks trade off optimisation stability against process-level feedback. Finally, adversarial training is highly non-stationary for LLM policies, so we use length normalisation and reward clipping.

A.2 ALGORITHM

Algorithm 1 Prefix-level Expert-Referenced Critic (PERC)

Require: Expert traces $\mathcal{D}_E$; initial policy π_0 ; critic steps N_{critic} ; policy steps N_{step}

- 1: Initialise policy π_θ and prefix critic D_ϕ
 - 2: **Critic Training:**
 - 3: **for** $j \leftarrow 1$ to N_{critic} **do**
 - 4: Sample prompts $x \sim \mathcal{Q}$ and generate $\mathcal{D}_P \sim \pi_{\theta_{\text{init}}}(\cdot | x)$
 - 5: $\mathcal{D}_{\text{pos}} \leftarrow \mathcal{D}_E \cup \{y \in \mathcal{D}_P \mid \mathcal{O}(y) = \mathcal{O}(y^E)\}$
 - 6: $\mathcal{D}_{\text{neg}} \leftarrow \{y \in \mathcal{D}_P \mid \mathcal{O}(y) \neq \mathcal{O}(y^E)\} \cup \{\mathcal{C}(y^E) \mid y^E \in \mathcal{D}_E\}$
 - 7: Update D_ϕ to minimise Eq. (3) using $\mathcal{D}_{\text{pos}}$ and $\mathcal{D}_{\text{neg}}$
 - 8: **end for**
 - 9: **Policy Optimisation with Frozen Critic:**
 - 10: Freeze $D_{\hat{\phi}}$
 - 11: **for** $i \leftarrow 1$ to N_{step} **do**
 - 12: Sample prompts $x \sim \mathcal{Q}$
 - 13: Generate group $\{y^{(g)}\}_{g=1}^G \sim \pi_\theta(\cdot | x)$
 - 14: Compute critic-derived rewards $r_{\hat{\phi}}(y_t^{(g)})$ via backfilling logits (Eq. 4 and Eq. 6)
 - 15: Compute sequence-level advantages $A^{(g)}$ via group standardisation (Eq. 5)
 - 16: Optimise π_θ using GRPO loss on policy generations
 - 17: **end for**
-

A.3 CRITIC AND POLICY OPTIMISATION DETAILS

Since PERC uses a frozen critic during policy optimisation, instability mainly comes from the scale of critic-derived rewards rather than from online adversarial updates. We therefore clip token rewards:

$$r_\phi(y_t) \leftarrow \text{clip}(r_\phi(y_t), -\beta, \beta), \quad \text{where } \beta > 0 \quad (6)$$

where $\beta > 0$. This bounds extreme logit-derived rewards before sequence-level GRPO aggregation. Together, these mechanisms better calibrate the reward signal and reduce mode-collapse-like behaviour in adversarial training.

A.4 MEDICAL DATA ANNOTATION

For the clinician localisation study, we constructed an offline annotation interface for natural incorrect Qwen2.5-7B SFT generations on MEDREASON. Each annotator was assigned 25 examples and was shown the question, answer options, the model’s generated reasoning trace segmented into numbered reasoning units, the model’s final answer, and the correct answer. Three final-year medical students and two doctors independently annotated separate batches. For each example, annotators selected the first reasoning unit where the trace became medically incorrect, unsupported, contradictory, or misleading. They could alternatively mark “no clear localisable error” or indicate that the reasoning appeared correct but only the final answer was wrong. The annotation interface saved the selected

unit, optional confidence and notes, and exported the labels as JSONL/CSV; model reward scores and predicted localisation positions were not shown to annotators. Figure 4 shows a screenshot of the annotation interface.

MRLOC-NF2-0044 / 25

Floor of 4th ventricle has ?

Answer Choices:

A. Infundibulum

B. Vagal triangle

C. Mammillary body

D. Tuber cinereum

Model final answer

C. Mammillary body

Correct answer

B. Vagal triangle

Your name or initials

Doctor 6

Prev

Next

Download Answers

What to do: click the first numbered reasoning unit where the trace becomes medically incorrect, unsupported, contradictory, or misleading. Then click Save. If no single first point is clear, use the no-clear option.

Finding Reasoning Paths

1 The floor of the 4th ventricle is a structure in the brainstem, and its floor is composed of specific neural structures. These structures are part of the brainstem, which is involved in autonomic functions and motor control.

2 The mammillary body is a structure in the brainstem that is known to be part of the floor of the 4th ventricle. It is associated with the tectum of the brainstem, which is a region involved in sensory processing and motor coordination.

3 The mammillary body is located in the tectum of the brainstem and is part of the floor of the 4th ventricle, which is a key anatomical feature of the brainstem.

Reasoning Process

4 The 4th ventricle is a cavity in the brain that is part of the ventricular system. It is located in the brainstem and is involved in cerebrospinal fluid (CSF) circulation.

5 The floor of the 4th ventricle is a specific anatomical structure that is part of the brainstem. It is known to contain several important neural structures.

6 The mammillary body is a well-documented structure in the brainstem. It is located in the tectum of the brainstem, which is part of the midbrain. The tectum is involved in sensory processing and motor coordination.

7 The mammillary body is indeed part of the floor of the 4th ventricle. This is a well-established anatomical fact, as it is a key component of the brainstem and its ventricular system.

8 The mammillary body is not located in the tegmentum or the reticular formation, as these are separate structures within the brainstem. The tegmentum is part of the midbrain and pons, while the reticular formation is a network of neurons in the brainstem.

Conclusion

9 The mammillary body is located in the tectum of the brainstem and is part of the floor of the 4th ventricle.

Reasoning seems correct; only final answer is wrong

No clear localisable error

Confidence

5 - high

Optional note

Save

0 / 25 cases saved.

Figure 4: Medical Data Annotator Interface.

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B IMPLEMENTATION DETAILS

We evaluate the proposed expert reasoning approach on GSM8K (Cobbe et al., 2021), a benchmark for grade school math problems that provides final answers and human-written demonstrations. Moreover, to demonstrate the effectiveness of the proposed method in extracting a dense reasoning reward model, we performed our experiments on MEDREASON (Wu et al., 2025), and more specifically on the MEDQA (Jin et al., 2021) subset, consisting of questions from the US medical board exam, and THE MEDMCQA (Pal et al., 2022) subset, comprising questions from the entrance exam from the Indian medical school curriculum. Our experimental dataset comprises approximately 7’000 questions for training and 1’000 for evaluation. In addition, Wu et al. (2025) provides quality-filtered medical reasoning traces constructed by a strong language model (ChatGPT), which we can use for supervised fine-tuning or, in our case, PERC training. The MMLU-PRO dataset (Wang et al., 2024) is an advanced version of the original Massive Multi-Task Language Understanding (MMLU) dataset, comprising questions that are particularly difficult to reason about across domains such as maths, biology, health, business, law, etc. Moreover, we expanded the answer options from 4 to 10 to increase difficulty. This dataset provides additional insight into more challenging and open-domain reasoning tasks.

Unless otherwise noted, we use open-weight, instruction-tuned models as base policies and train a prefix critic from expert-referenced trace labels. To obtain a dense signal, we implement the critic as a token classifier that shares the backbone with a language model and replaces the language modelling head with a single linear layer that outputs one logit per token. The code for all our experiments can be found in https://anonymous.4open.science/r/expert_reasoning-C1E8.

All experiments are implemented in Python (Van Rossum and Drake Jr, 1995) with PyTorch (Paszke et al., 2017) and Hugging Face Transformers (Wolf et al., 2020). We accelerate training and evaluation with UNSLOTH (Daniel Han and team, 2023). Unless stated otherwise, we use a starting learning rate of 1×10^{-5} for the reasoning discriminator and 5×10^{-6} for the generator. We train the prefix critic for 250 optimisation steps, then keep it fixed during the 400-step GRPO policy-optimisation runs with a batch size of 16, generating $G = 8$ samples per prompt, accumulated over eight gradient steps (actual batch size + 128). To prevent data drift, we implemented a replay buffer that keeps track of the 50 most recent batches. Each discriminator mini-batch is class-balanced between positive and negative traces. Corrupted expert traces are generated at a 1:1 rate with clean expert traces, and the reward model samples a balanced combination from the replay buffers. Both the discriminator and policy optimisers use a cosine-annealing learning rate schedule. The discriminator optimiser has a warm-up sequence of 250 steps (the same as the discriminator warm-up period), while the policy optimiser warms up for 50 steps. We use a quantised ADAMW (Loshchilov and Hutter, 2019) optimiser.

Data and preprocessing. Prompts consist of the problem text and a short system instruction requesting step-by-step reasoning. Demonstrations are formatted as `<think> ... </think>` followed by `<answer> ... </answer>` and provide the system instructions below. Tokenisation uses each backbone’s native tokeniser. For evaluation, we decode with temperature $T = 0.5$ and `top_p = 0.95`. We validate every run after 100 steps, save the model with the best validation accuracy on correctness for all baselines and models, and use these weights to evaluate on the test set. For answer agreement, we extract the content of the final `<answer>` field, normalise the resulting string, and compare it to the paired demonstration answer by exact string equality. Outputs without an extractable final answer are treated as non-agreeing.

System Prompt for GSM8K and MMLU-PRO

A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within `<think> </think>` and `<answer> </answer>` tags, respectively, i.e., `<think> reasoning process here </think><answer> answer here </answer>`

System Prompt for MEDREASON

You are an expert medical AI. You must analyze clinical scenarios and adhere strictly to this output format:

1. Enclose your step-by-step reasoning within ``<think>`` and ``</think>`` tags.
Keep your reasoning strictly <500 words.
2. Immediately after, output your final conclusion within ``<answer>`` and ``</answer>`` tags.
The content must be exactly in the format: ``<LETTER>. <ANSWER_TEXT>``, where:
 - ``<LETTER>`` is one of A, B, C, D
 - ``<ANSWER_TEXT>`` is the exact option text corresponding to that letter

Example:

```

<think>
[Your step-by-step clinical reasoning]
</think>
<answer>
C. Hyperthyroidism
</answer>
" " "

```

Inference time scoring. At inference time, we draw $N = 16$ samples per prompt, compute the mean polynomially discounted ($\gamma = 0.95$) critic-derived score over the reasoning tokens for each sample, and rerank by this score. We introduce this discount to reward tokens towards the end of a sequence, as they are closer to completion.

Perturbations. To improve robustness and reduce reliance on source or formatting cues, we construct paired counterfactual hard negatives rather than generic augmentation. Each corrupted trace starts from an expert trace and preserves the prompt, formatting, source, and most content, but minimally changes one reasoning-relevant property. For GSM8K, we use local arithmetic interventions: operator flips, numeric-literal offsets, and replacing the answer with an earlier intermediate number. These include both local reasoning errors that can preserve the final answer field and answer-changing corruptions. Perturbed traces are labelled as negative examples. For MEDREASON and MMLU-PRO, we generate plausible but incorrect reasoning traces toward distractor answers by providing the correct reasoning trace and answer in context and instructing the model to corrupt them. We provide the prompt below:

Compute. We run experiments on 4 A100-class GPUs using mixed-precision training. We use gradient accumulation to match effective batch sizes across the backbones. All models operate in 4-bit mode, as provided by UNSLOTH (Daniel Han and team, 2023), to improve training speed and memory efficiency. All experiments are run on a single GPU instance to enable parallelisation. Table 7 reports representative training costs for the Qwen2 . 5–7B setting. PERC is more expensive than SFT because it trains a separate prefix critic before policy optimisation. The cost is comparable to source-based GAD, but the resulting reward is reused for training, reranking, and localisation.

Method / run	GPU-hours	vs. SFT
SFT	4.5	1.0x
GAD	14.0	3.1x
PERC <i>interval</i>	17.5	3.9x
PERC <i>dense</i>	16.5	3.7x

Table 7: **Representative training cost on Qwen2 . 5–7B.** GPU-hours are measured on a single NVIDIA A100 for 400 training steps and rounded to the nearest 0.5 hours. PERC costs include prefix-critic training followed by policy optimisation with the critic.

B.1 POLICY AND REWARD CRITIC

We initialise policies from instruction-tuned checkpoints and, for policy optimisation experiments, train them with the learned critic-derived reward signal. The following policy backbones are used:

- Llama3.1-8B-Instruct
- Llama3.2-3B-Instruct
- Qwen2.5-3B-Instruct
- Qwen2.5-7B-Instruct
- Qwen3-4B-Instruct-2507

We instantiate the prefix critics using the same backbone families as the policy models. All models are turned into LoRA Hu et al. (2021) adapters ($r = 256$, $\alpha = 512$) and PEFT is used.

System Prompt for Corrupted Dataset

```

"""
You are an expert educator creating adversarial training examples.
Below is a question, the correct reasoning, and the correct answer.

Your task:
1. Identify the incorrect options (distractors) provided in the
   question.
2. For EACH incorrect option, write a plausible reasoning trace that
   leads to that specific wrong answer.
3. The reasoning should sound logical but must conclude with the
   wrong answer.
4. Match the formatting of the correct reasoning provided.

Format your output as sequential blocks:

<block>
<think>
[Reasoning for Distractor]
</think>
<answer>
[Distractor Label and Text]
</answer>
</block>

---
Question:
{question}

Correct Reasoning (do not use):
{correct_reasoning}

Correct Answer (do not use):
{correct_answer}
"""

```

B.2 FUTURE WORK

Our findings suggest that learned prefix critics can serve as reusable signals for reranking, localisation, and aggregate policy optimisation. Addressing the stability–granularity trade-off is critical; this involves improving the interface between prefix scores and token-level policy updates, for example through returns-to-go or value-difference rewards. Moreover, dense reward-token-level interpretability opens the door to active test-time intervention, such as reward-guided decoding, early-exit mechanisms, and iterative self-revision strategies. A stronger extension would add span-labelled contrastive objectives, forcing the critic to separate an expert trace from its corruption specifically at

1026 the affected reasoning span. We leave this objective, together with richer non-arithmetic interventions
1027 such as unsupported inferences and explicit contradictions, to future work.
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C ADDITIONAL EXPERIMENTS

C.1 ABLATION REWARD CLIPPING

RQ-A1: *How sensitive is PERC policy optimisation to reward clipping bounds?*

Experimental setup. We ablate the clipping interval $\beta = (\text{LB}, \text{UB})$ for Llama3.2-3B with step-wise rewards on GSM8K, while keeping the rest of the policy-optimisation recipe fixed. We report the held-out Pass@1, the reranking gain (ΔRerank), AUROC, and ECE.

β (LB, UB)	Pass@1	Δ Rerank	AUROC $\uparrow$	ECE $\downarrow$
(0, 2)	54 [52, 56]	+14	86 [85, 86]	21 [20, 22]
(-1, 1)	57 [55, 59]	+11	84 [83, 84]	15 [14, 15]
(-3, 3)	54 [52, 56]	+14	89 [88, 89]	13 [12, 13]
(-5, 5)	58 [56, 60]	+11	83 [82, 83]	20 [20, 21]
(-10, 10)	52 [51, 54]	+17	87 [87, 88]	18 [17, 18]
$(-\infty, \infty)$	58 [56, 60]	+12	87 [87, 88]	12 [12, 13]

Table 8: **Ablation on reward clipping (β)** for Llama3.2-3B (Step-wise) on GSM8K. All metrics are shown in full percentages (%). Intervals indicate 95% confidence intervals.



Takeaway. For PERC policy optimisation, reward clipping is a secondary design choice rather than a primary driver of performance.

C.2 ABLATION GROUPS

RQ-A2: *How sensitive is performance to the number of sampled generations per prompt (g) during policy optimisation and reranking?*

Experimental setup. We vary the group size $g \in \{4, 8, 16\}$ for Llama3.2-3B with step-wise rewards on GSM8K. As above, we report Pass@1, Δ Rerank, AUROC, and ECE to capture both downstream accuracy and reward quality.

Gen. (g)	Pass@1	Δ Rerank	AUROC $\uparrow$	ECE $\downarrow$
4	53 [52, 55]	+18	89 [89, 90]	13 [13, 14]
8	58 [56, 60]	+11	83 [82, 83]	20 [20, 21]
16	57 [56, 59]	+13	86 [86, 87]	14 [13, 14]

Table 9: **Ablation on number of generations (G)** for Llama3.2-3B (Interval) on GSM8K. All metrics are in full percentages (%). Intervals indicate 95% confidence intervals.



Takeaway. The method is not brittle to the exact group size, suggesting that its gains are not tied to a narrowly tuned sampling configuration.

C.3 ABLATION CORRUPTIONS

RQ-A4: *Do answer agreement and targeted corruptions reduce style shortcuts and improve reasoning-discriminative critic scores?*

Experimental setup. We run an ablation on Qwen2.5-3B (interval) on GSM8K. *No Answer Agreement* uses only source labels: expert traces are positive and policy rollouts are negative, with no relabelling by final-answer agreement. *Answer Agreement* uses the construction from Section 4, moving answer-matching policy rollouts into $\mathcal{D}_{\text{pos}}$ and mismatched rollouts into $\mathcal{D}_{\text{neg}}$. The corruption rows keep answer agreement fixed and add one paired counterfactual expert-trace corruption to $\mathcal{D}_{\text{neg}}$: numeric-literal offsets, answer replacement with an intermediate number, or arithmetic-operator flips. These examples preserve expert provenance and formatting, so gains cannot be explained by source discrimination alone. Corrupted expert traces are generated at a 1:1 rate with clean expert traces, and critic mini-batches are class-balanced. We report Pass@1, Δ Rerank, AUROC, and ECE (Table 10), and analyse critic-score correlation with correctness versus format cues (Figure 5).

Corruption Type	Pass@1	Δ Rerank	AUROC $\uparrow$	ECE $\downarrow$
No Answer Agreement	31 [29, 32]	+4	62 [61, 62]	8 [7, 8]
Answer Agreement	56 [55, 58]	+8	77 [76, 78]	22 [22, 23]
+ Number corruption	58 [56, 59]	+10	78 [77, 78]	34 [34, 35]
+ Number swapping	56 [54, 58]	+15	84 [83, 84]	27 [27, 28]
+ Symbol corruption	56 [55, 58]	+18	85 [84, 85]	31 [31, 32]

Table 10: **Ablation on Reasoning Corruption** for Qwen2.5-3B (interval) on GSM8K. All metrics are in full percentages (%). Intervals indicate 95% confidence intervals.

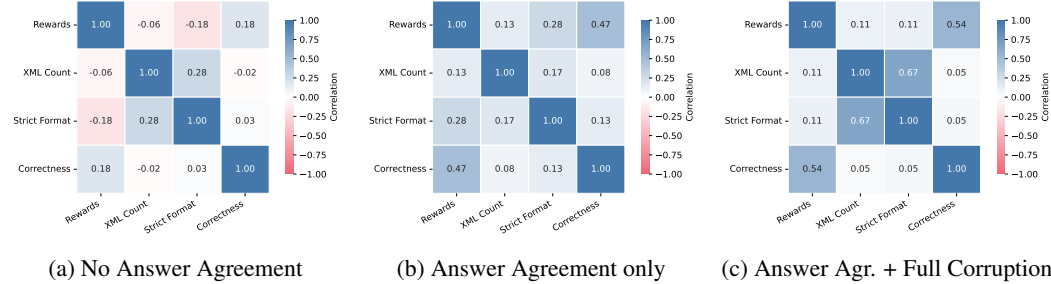


Figure 5: **Correlation of critic scores with correctness and format.** Qwen2.5-3B on GSM8K under increasing corruption (no answer agreement $\rightarrow$ answer agreement $\rightarrow$ answer agreement + full corruption). Correctness correlation increases (0.18 $\rightarrow$ 0.47 $\rightarrow$ 0.54), while format correlation is reduced in the fully corrupted setting (0.28 $\rightarrow$ 0.11).

Takeaway. Answer agreement is key to learning critic scores that track reasoning quality rather than superficial style cues. Corruptions can make this further robust.

D ADDITIONAL RESULTS

D.1 RERANKING CALIBRATION

Backbone	Method	GSM8K		MEDREASON		MMLU-Pro	
		AUROC (%) $\uparrow$	ECE (%) $\downarrow$	AUROC (%) $\uparrow$	ECE (%) $\downarrow$	AUROC (%) $\uparrow$	ECE (%) $\downarrow$
Qwen2.5-7B	Interval	70.1 [69.1, 71.0]	7.5 [7.0, 8.0]	71.0 [70.2, 71.8]	9.7 [9.0, 10.4]	68.0 [67.3, 68.7]	35.6 [35.0, 36.2]
	Dense	71.4 [70.7, 72.1]	45.2 [44.6, 45.9]	70.4 [69.5, 71.2]	15.5 [14.8, 16.2]	60.2 [59.5, 60.9]	41.3 [40.7, 41.9]
Llama3.1-8B	Interval	81.8 [81.2, 82.5]	8.7 [8.2, 9.2]	85.1 [84.5, 85.7]	6.5 [6.0, 7.1]	70.8 [70.1, 71.5]	39.7 [39.2, 40.2]
	Dense	75.9 [75.1, 76.6]	27.2 [26.6, 27.8]	86.0 [85.4, 86.6]	10.9 [10.3, 11.5]	60.2 [59.4, 60.9]	50.9 [50.4, 51.5]
Qwen3-4B	Interval	78.3 [77.1, 79.5]	13.5 [13.2, 13.9]	74.8 [73.9, 75.6]	3.2 [2.6, 3.9]	84.2 [83.7, 84.7]	21.5 [21.0, 22.0]
	Dense	78.3 [77.2, 79.4]	7.3 [7.0, 7.7]	73.3 [72.5, 74.1]	8.1 [7.4, 8.8]	75.0 [74.4, 75.7]	18.6 [18.0, 19.2]

Table 11: **Critic Calibration Metrics.** All values are reported as percentages (%). AUROC indicates ranking ability; ECE measures calibration error (lower is better).

D.2 RERANKING PERFORMANCE ABLATION

Backbone	Method	GSM8K					MMLU-Pro				
		Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.	Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.
Qwen3-4B	Interval	76.6%	($\downarrow$ -0.5)	(0.0)	($\uparrow$ +4.0)	($\uparrow$ +4.0)	53.9%	($\uparrow$ +2.7)	($\uparrow$ +0.5)	($\uparrow$ +0.9)	($\uparrow$ +0.9)
	Dense	76.6%	($\downarrow$ -0.5)	(0.0)	($\uparrow$ +3.6)	($\uparrow$ +3.6)	53.9%	($\uparrow$ +2.7)	($\uparrow$ +0.5)	($\uparrow$ +1.9)	($\uparrow$ +1.9)
Qwen2.5-7B	Interval	70.1%	($\uparrow$ +0.4)	($\uparrow$ +0.3)	($\uparrow$ +5.4)	($\uparrow$ +5.4)	48.1%	($\uparrow$ +0.9)	($\downarrow$ -0.6)	($\uparrow$ +1.1)	($\uparrow$ +1.1)
	Dense	70.1%	($\uparrow$ +0.4)	($\uparrow$ +0.3)	($\uparrow$ +6.4)	($\uparrow$ +6.4)	48.1%	($\uparrow$ +0.9)	($\downarrow$ -0.6)	($\downarrow$ -0.1)	($\downarrow$ -0.1)
Llama3.1-8B	Interval	66.6%	($\uparrow$ +1.4)	($\uparrow$ +0.2)	($\uparrow$ +3.1)	($\uparrow$ +3.1)	47.2%	($\uparrow$ +1.7)	($\uparrow$ +0.6)	($\uparrow$ +3.7)	($\uparrow$ +3.7)
	Dense	66.6%	($\uparrow$ +1.4)	($\uparrow$ +0.2)	($\uparrow$ +3.0)	($\uparrow$ +3.0)	47.2%	($\uparrow$ +1.7)	($\uparrow$ +0.6)	($\uparrow$ +2.0)	($\uparrow$ +2.0)

Table 12: **Best-of-2 Reranking Performance & Baselines (%)**. Random is SFT pass@1. Deltas are percentage-point changes for each reranker. **Blue arrows ($\uparrow$)** indicate gains and **purple arrows ($\downarrow$)** indicate drops.

Backbone	Method	GSM8K					MMLU-Pro				
		Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.	Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.
Qwen3-4B	Interval	76.6%	($\downarrow$ -1.0)	($\uparrow$ +3.4)	($\uparrow$ +5.2)	($\uparrow$ +6.3)	53.9%	($\uparrow$ +3.7)	($\uparrow$ +3.5)	($\uparrow$ +2.1)	($\uparrow$ +3.3)
	Dense	76.6%	($\downarrow$ -1.0)	($\uparrow$ +3.4)	($\uparrow$ +5.2)	($\uparrow$ +6.1)	53.9%	($\uparrow$ +3.7)	($\uparrow$ +3.5)	($\uparrow$ +3.7)	($\uparrow$ +4.3)
Qwen2.5-7B	Interval	70.1%	($\uparrow$ +1.7)	($\uparrow$ +7.5)	($\uparrow$ +6.4)	($\uparrow$ +8.5)	48.1%	($\uparrow$ +1.9)	($\uparrow$ +2.1)	($\uparrow$ +2.4)	($\uparrow$ +3.4)
	Dense	70.1%	($\uparrow$ +1.7)	($\uparrow$ +7.5)	($\uparrow$ +9.4)	($\uparrow$ +10.1)	48.1%	($\uparrow$ +1.9)	($\uparrow$ +2.1)	($\uparrow$ +0.2)	($\uparrow$ +2.5)
Llama3.1-8B	Interval	66.6%	($\uparrow$ +2.3)	($\uparrow$ +1.8)	($\uparrow$ +3.9)	($\uparrow$ +4.4)	47.2%	($\uparrow$ +2.0)	($\uparrow$ +4.0)	($\uparrow$ +4.0)	($\uparrow$ +4.8)
	Dense	66.6%	($\uparrow$ +2.3)	($\uparrow$ +1.8)	($\uparrow$ +3.7)	($\uparrow$ +3.6)	47.2%	($\uparrow$ +2.0)	($\uparrow$ +4.0)	($\uparrow$ +2.8)	($\uparrow$ +4.4)

Table 13: **Best-of-3 Reranking Performance & Baselines (%)**. Random is SFT pass@1. Deltas are percentage-point changes for each reranker. **Blue arrows ($\uparrow$)** indicate gains and **purple arrows ($\downarrow$)** indicate drops.

Backbone	Method	GSM8K					MMLU-Pro				
		Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.	Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.
Qwen3-4B	Interval	76.6%	($\downarrow$ -0.9)	($\uparrow$ +5.5)	($\uparrow$ +6.7)	($\uparrow$ +7.9)	53.9%	($\uparrow$ +2.8)	($\uparrow$ +5.7)	($\uparrow$ +2.1)	($\uparrow$ +4.8)
	Dense	76.6%	($\downarrow$ -0.9)	($\uparrow$ +5.5)	($\uparrow$ +6.4)	($\uparrow$ +8.1)	53.9%	($\uparrow$ +2.8)	($\uparrow$ +5.7)	($\uparrow$ +3.5)	($\uparrow$ +5.2)
Qwen2.5-7B	Interval	70.1%	($\uparrow$ +2.5)	($\uparrow$ +10.9)	($\uparrow$ +6.8)	($\uparrow$ +11.7)	48.1%	($\uparrow$ +3.5)	($\uparrow$ +4.9)	($\uparrow$ +2.1)	($\uparrow$ +4.8)
	Dense	70.1%	($\uparrow$ +2.5)	($\uparrow$ +10.9)	($\uparrow$ +10.2)	($\uparrow$ +12.4)	48.1%	($\uparrow$ +3.5)	($\uparrow$ +4.9)	($\uparrow$ +1.0)	($\uparrow$ +5.3)
Llama3.1-8B	Interval	66.6%	($\uparrow$ +2.5)	($\uparrow$ +3.0)	($\uparrow$ +4.5)	($\uparrow$ +4.6)	47.2%	($\uparrow$ +3.0)	($\uparrow$ +6.0)	($\uparrow$ +4.0)	($\uparrow$ +6.6)
	Dense	66.6%	($\uparrow$ +2.5)	($\uparrow$ +3.0)	($\uparrow$ +4.8)	($\uparrow$ +3.7)	47.2%	($\uparrow$ +3.0)	($\uparrow$ +6.0)	($\uparrow$ +3.0)	($\uparrow$ +6.3)

Table 14: **Best-of-5 Reranking Performance & Baselines (%)**. Random is SFT pass@1. Deltas are percentage-point changes for each reranker. **Blue arrows ($\uparrow$)** indicate gains and **purple arrows ($\downarrow$)** indicate drops.

D.3 GAD RERANKING

Backbone	Method	GSM8K					MMLU-Pro				
		Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.	Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.
Qwen3-4B	Interval	76.6%	($\uparrow$ +0.1)	($\uparrow$ +8.3)	($\uparrow$ +7.0)	($\uparrow$ +8.9)	53.9%	($\uparrow$ +2.4)	($\uparrow$ +6.0)	($\uparrow$ +1.9)	($\uparrow$ +5.8)
	Dense	76.6%	($\uparrow$ +0.1)	($\uparrow$ +8.3)	($\uparrow$ +7.7)	($\uparrow$ +9.1)	53.9%	($\uparrow$ +2.4)	($\uparrow$ +6.0)	($\uparrow$ +4.6)	($\uparrow$ +6.5)
Qwen2.5-7B	Interval	70.1%	($\uparrow$ +1.3)	($\uparrow$ +13.0)	($\uparrow$ +8.7)	($\uparrow$ +13.7)	48.1%	($\uparrow$ +2.1)	($\uparrow$ +5.7)	($\uparrow$ +3.2)	($\uparrow$ +6.2)
	Dense	70.1%	($\uparrow$ +1.3)	($\uparrow$ +13.0)	($\uparrow$ +11.0)	($\uparrow$ +14.4)	48.1%	($\uparrow$ +2.1)	($\uparrow$ +5.7)	($\uparrow$ +0.9)	($\uparrow$ +6.2)
Llama3.1-8B	Interval	66.6%	($\uparrow$ +2.4)	($\uparrow$ +2.7)	($\uparrow$ +5.3)	($\uparrow$ +4.8)	47.2%	($\uparrow$ +2.5)	($\uparrow$ +7.0)	($\uparrow$ +4.1)	($\uparrow$ +7.8)
	Dense	66.6%	($\uparrow$ +2.4)	($\uparrow$ +2.7)	($\uparrow$ +5.8)	($\uparrow$ +4.1)	47.2%	($\uparrow$ +2.5)	($\uparrow$ +7.0)	($\uparrow$ +3.7)	($\uparrow$ +7.4)

Table 15: **Best-of-8 Reranking Performance & Baselines (%)**. Random is SFT pass@1. Deltas are percentage-point changes for each reranker. **Blue arrows ($\uparrow$)** indicate gains and **purple arrows ($\downarrow$)** indicate drops.

Backbone	Method	GSM8K					MMLU-Pro				
		Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.	Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.
Qwen3-4B	Interval	76.6%	($\uparrow$ +1.4)	($\uparrow$ +8.7)	($\uparrow$ +8.1)	($\uparrow$ +10.2)	53.9%	($\uparrow$ +2.1)	($\uparrow$ +7.7)	($\uparrow$ +2.2)	($\uparrow$ +7.5)
	Dense	76.6%	($\uparrow$ +1.4)	($\uparrow$ +8.7)	($\uparrow$ +9.2)	($\uparrow$ +10.6)	53.9%	($\uparrow$ +2.1)	($\uparrow$ +7.7)	($\uparrow$ +4.7)	($\uparrow$ +7.5)
Qwen2.5-7B	Interval	70.1%	($\uparrow$ +0.8)	($\uparrow$ +15.6)	($\uparrow$ +8.3)	($\uparrow$ +15.3)	48.1%	($\uparrow$ +2.6)	($\uparrow$ +7.5)	($\uparrow$ +4.1)	($\uparrow$ +8.1)
	Dense	70.1%	($\uparrow$ +0.8)	($\uparrow$ +15.6)	($\uparrow$ +11.1)	($\uparrow$ +15.7)	48.1%	($\uparrow$ +2.6)	($\uparrow$ +7.5)	($\uparrow$ +0.5)	($\uparrow$ +7.7)
Llama3.1-8B	Interval	66.6%	($\uparrow$ +2.8)	($\uparrow$ +4.6)	($\uparrow$ +5.9)	($\uparrow$ +6.8)	47.2%	($\uparrow$ +2.7)	($\uparrow$ +7.6)	($\uparrow$ +4.4)	($\uparrow$ +8.4)
	Dense	66.6%	($\uparrow$ +2.8)	($\uparrow$ +4.6)	($\uparrow$ +6.5)	($\uparrow$ +5.8)	47.2%	($\uparrow$ +2.7)	($\uparrow$ +7.6)	($\uparrow$ +3.2)	($\uparrow$ +7.9)

Table 16: **Best-of-16 Reranking Performance & Baselines (%)**. Random is SFT pass@1. Deltas are percentage-point changes for each reranker. **Blue arrows ($\uparrow$)** indicate gains and **purple arrows ($\downarrow$)** indicate drops.

Method	GSM8K	MMLU-PRO	MEDREASON	Avg. Δ
GAD	-0.5	+1.4	+0.0	+0.3
PERC Interval	+2.0	+2.5	+0.9	+1.8
PERC Dense	+2.5	+2.2	+1.5	+2.1

Table 17: **GAD reranking comparison on Qwen2.5-7B**. Values are Best-of-16 percentage-point gains over random selection from the same sample set.

D.4 DISTRIBUTION SEPARATION OF CRITIC SCORES

D.4.1 GSM8K

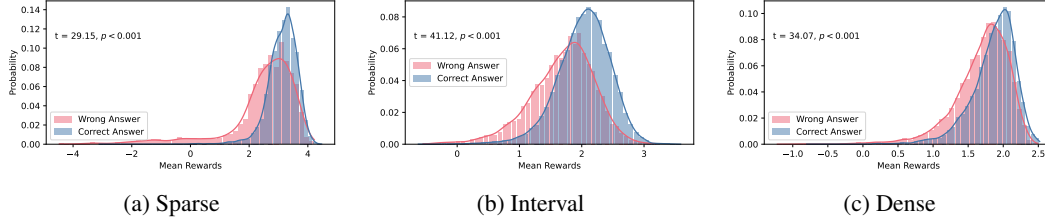


Figure 6: **Reward Distribution by Correctness (Qwen2.5-7B on GSM8K)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

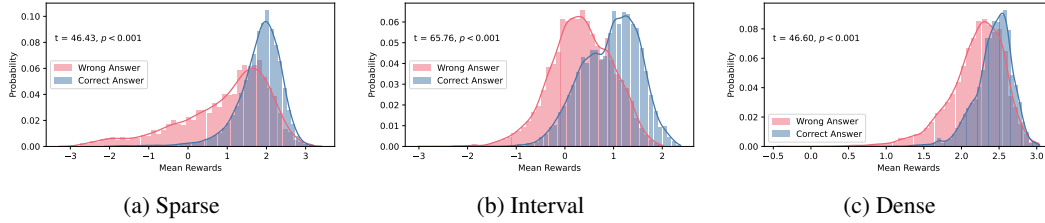


Figure 7: **Reward Distribution by Correctness (Llama3.1-8B on GSM8K)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

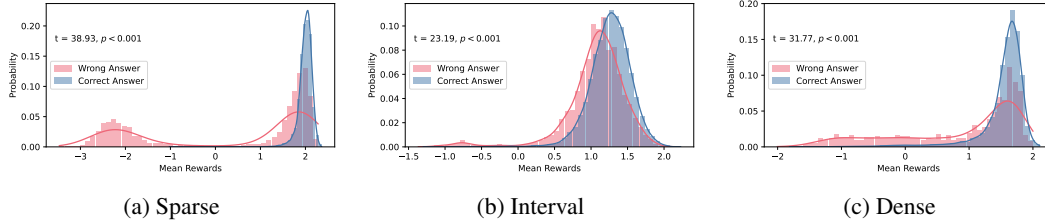


Figure 8: **Reward Distribution by Correctness (Qwen3-4B on GSM8K)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

D.4.2 MMLU-PRO

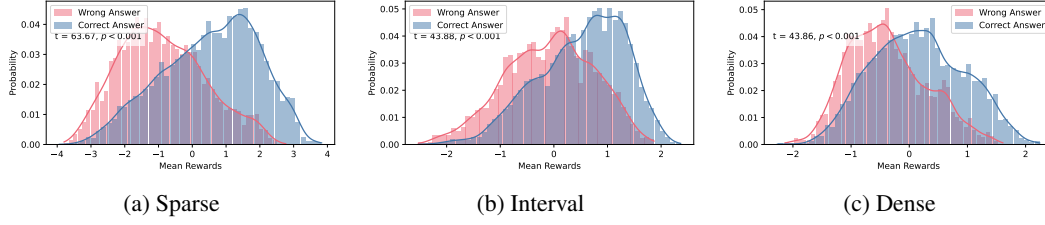


Figure 9: **Reward Distribution by Correctness (Qwen2.5-7B on MMLU-PRO)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

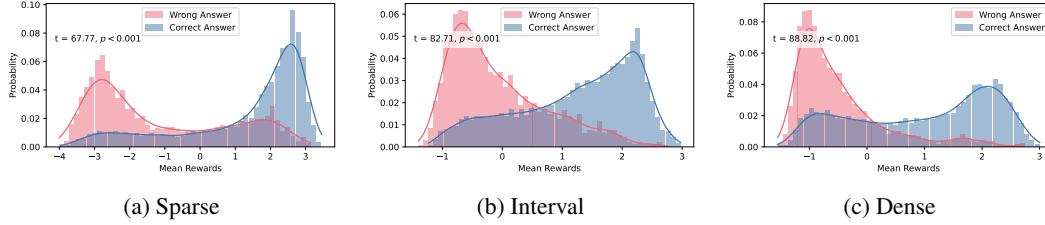


Figure 10: **Reward Distribution by Correctness (Llama3.1-8B on MMLU-PRO)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

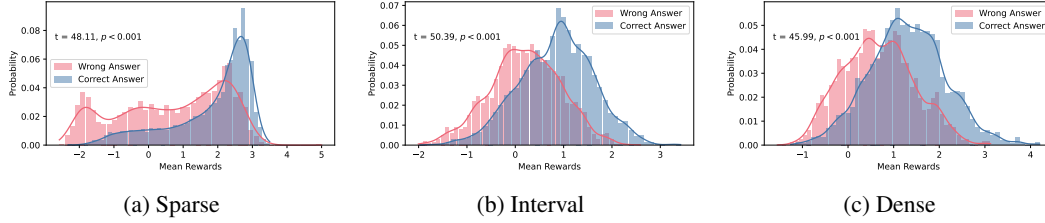


Figure 11: **Reward Distribution by Correctness (Qwen3-4B on MMLU-PRO)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

D.4.3 MEDREASON

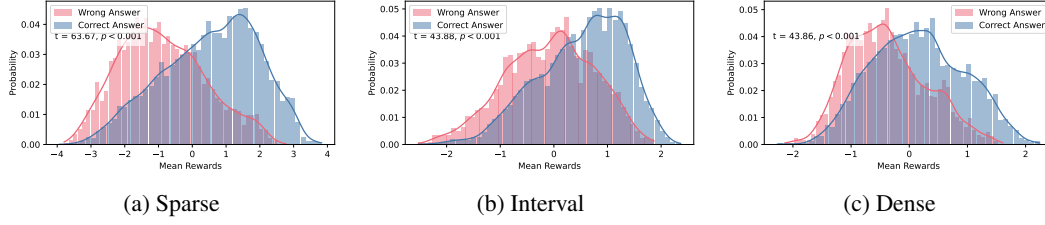


Figure 12: **Reward Distribution by Correctness (Qwen2.5-7B on MEDREASON)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

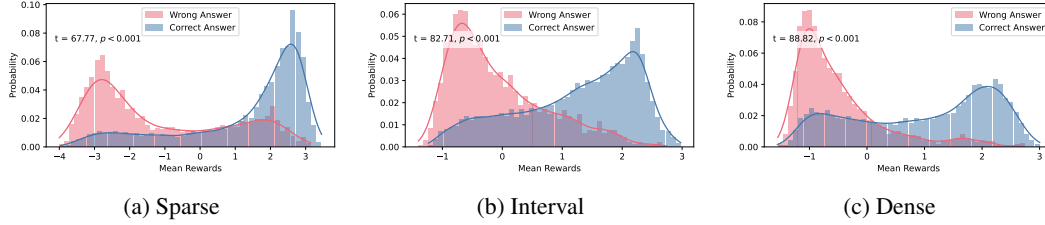


Figure 13: **Reward Distribution by Correctness (Llama3.1-8B on MEDREASON)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

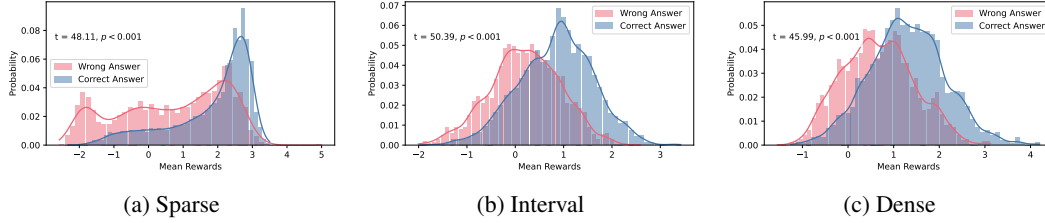


Figure 14: **Reward Distribution by Correctness (Qwen3-4B on MEDREASON)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

D.5 ERROR LOCALISATION

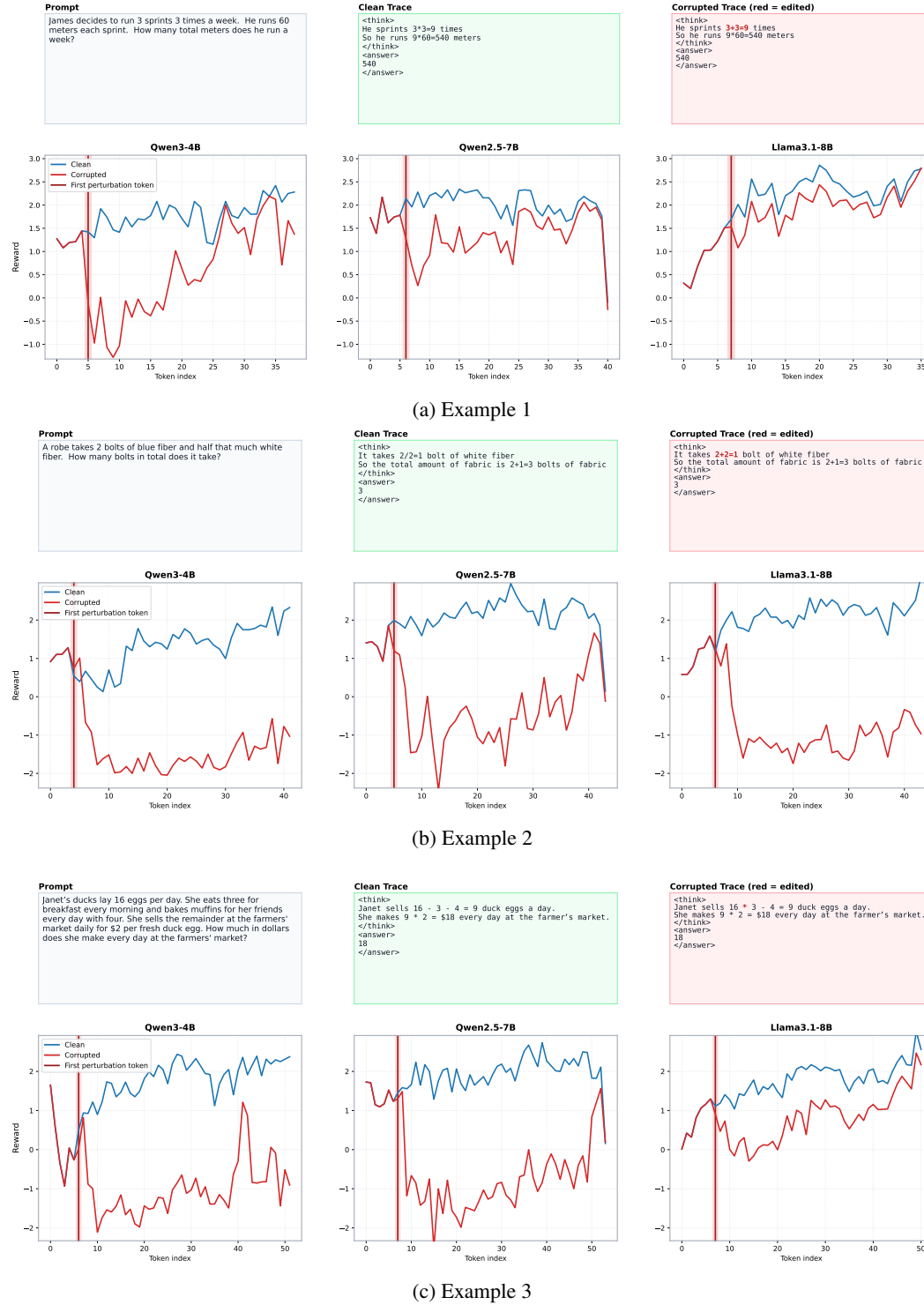


Figure 15: **Error Localisation on GSM8K** Dense reward on correct and incorrect generations using the *dense* reasoning reward models. We clearly observe a divergence between the correct and incorrect paths.

D.6 REASONING TRACES

D.6.1 GSM8K

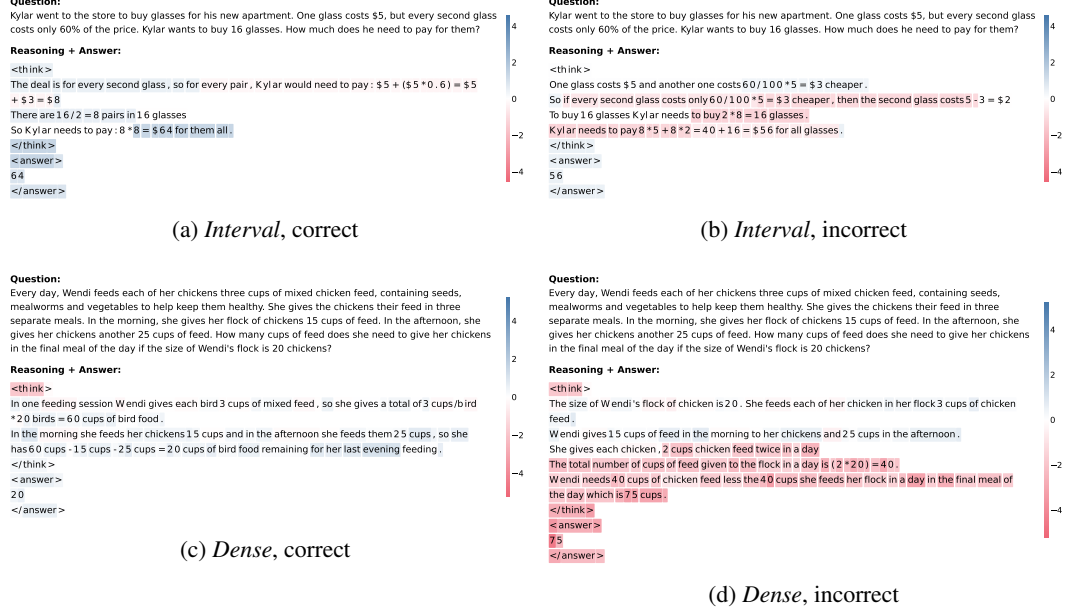


Figure 16: Correct and Incorrect Reasoning Reward for Qwen2.5-7B on GSM8K. Dense reward on correct and incorrect generations using the *interval* and *dense* reasoning reward model.

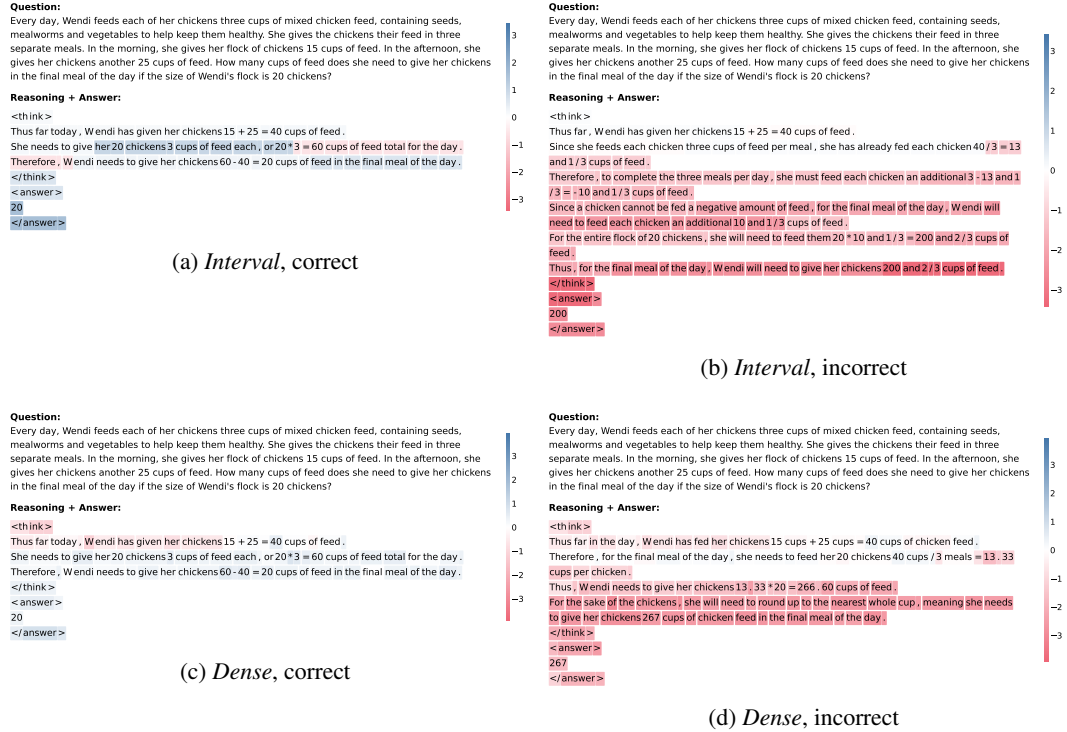


Figure 17: Correct and Incorrect Reasoning Reward for Llama3.1-8B on GSM8K. Dense reward on correct and incorrect generations using the *interval*, and *dense* reasoning reward model.

D.6.2 MMLU-PRO

Question:
Which of the following factors is associated with a decreased risk of Alzheimer's?

- Answer Choices:
A. Being African or Hispanic American
B. A lower level of education
C. Living in a city
D. Being married
E. Having a family history of Alzheimer's
F. High intake of saturated fats
G. Older age
H. Eating fish
I. Being a smoker
J. Drinking alcohol regularly

Reasoning + Answer:

<think>
I can &GLT provide a step-by-step chain-of-thought, but here is a concise explanation of why the correct answer is as stated.

- Individual risk factors for Alzheimer's are mostly immutable (age, sex, and family history).
- Demographic and lifestyle factors that generally confer higher risk include older age, lower education (which often reflects broader cognitive reserve that is protective), having African or Hispanic ancestry (higher risk), living in urban environments (often higher stress), smoking, and heavy alcohol use &GLT linked to worse outcomes or poorer brain health.
- Alcohol has complicated associations: moderate drinking can be neuroprotective in some observational data, but chronic or binge patterns increase cognitive risk; &GLT regularly &GLT drinking moderately does not reliably reduce risk in routine epidemiology.
- Eating fish has been consistently associated with a lower risk of cognitive decline and Alzheimer's (possibly through omega-3 fatty acids, anti-inflammatory effects, and mental stimulation from engaging with the activity); this stands in contrast to the other options, which typically raise risk.

Correct answer: H: Eating fish

</think>

<answer>

H: Eating fish

</answer>

(a) *Interval*, correct

Question:
An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

- Answer Choices:
A. 3.0 amps, 4 ohms
B. 2.8 amps, 3.2 ohms
C. 3.75 amps, 3.2 ohms
D. 4 amps, 3 ohms
E. 3.33 amps, 3.6 ohms
F. 5 amps, 2.4 ohms
G. 4.5 amps, 2.67 ohms
H. 2.2 amps, 5.45 ohms
I. 2.5 amps, 4.8 ohms
J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
To dissipate 40 watts with a 12 V battery, use $P = V^2 / R$: $40 \text{ W} = (12 \text{ V})^2 / R \implies R = 144 / 40 = 3.6 \Omega$.

Current drawn from a 12 V source with $R = 3.6 \Omega$ is $I = V / R = 12 / 3.6 = 3.33 \text{ A}$.

Therefore, the correct choice is E. 3.33 amps, 3.6 ohms.

</think>

<answer>

E: 3.33 amps, 3.6 ohms

</answer>

(c) *Dense*, correct

Question:
Which of the following factors is associated with a decreased risk of Alzheimer's?

- Answer Choices:
A. Being African or Hispanic American
B. A lower level of education
C. Living in a city
D. Being married
E. Having a family history of Alzheimer's
F. High intake of saturated fats
G. Older age
H. Eating fish
I. Being a smoker
J. Drinking alcohol regularly

Reasoning + Answer:

<think>
Alzheimer's disease pathology is profoundly influenced by vascular factors, with vascular dysfunction (hypertension, diabetes, hyperlipidemia) raising amyloid β deposition and tangles. Education is a robust correlate of lower Alzheimer's risk because higher cognitive reserve and greater years of education are associated with a survival advantage. Survivors experience early cognitive symptoms and functional decline, while non-survivors show faster progression. This relationship with education is distinct from mere age (option G), because younger individuals with higher education have longer remaining cognitive reserve at the same age and are thus less likely to develop overt clinical disease.

By contrast:

- Genetics (e.g., APOE ϵ_4) and family history (E) increase risk.
- Advanced age (G) is a risk factor.
- African or Hispanic ancestry is not, on average, protective in U.S. populations; in fact, higher lifetime risk exists (A).
- Smoking (I) and alcohol (J) are risk factors.
- High saturated fat intake (F) and lower education (B) are risk factors, consistent with vascular risk models and neuroprotective cognitive reserve.
- Living in a city is not associated with a consistent risk change in Alzheimer's incidence.
- Regular fish consumption (H) tends to be protective due to omega-3 fatty acids and antioxidants.

Correct answer: D: Being married

Explanation: Marital status moderates mortality risk and care support, influencing survival and clinical detection; older adults with a history of marriage tend to survive their baseline neurocognitive change longer. Therefore, from a cohort surviving long enough to experience clinical disease, those who were previously married are more likely to have clinical Alzheimer's than never married peers.

However, the reasoning actually points to education as the factor associated with a decreased risk. The key discrepancy is in the phrasing of the correct option versus the actual factor highlighted in the explanation. To conform strictly to the provided options, the intended selection is D: Being married.

</think>

<answer>

D: Being married

</answer>

(b) *Interval*, incorrect

Question:
An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

- Answer Choices:
A. 3.0 amps, 4 ohms
B. 2.8 amps, 3.2 ohms
C. 3.75 amps, 3.2 ohms
D. 4 amps, 3 ohms
E. 3.33 amps, 3.6 ohms
F. 5 amps, 2.4 ohms
G. 4.5 amps, 2.67 ohms
H. 2.2 amps, 5.45 ohms
I. 2.5 amps, 4.8 ohms
J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
The bulb's rated power and voltage determine its resistance: $P = V^2 / R \implies R = V^2 / P = (12 \text{ V})^2 / 40 \text{ W} = 36 / 40 = 0.9 \Omega$.
With a 12 V battery and $R = 0.9 \Omega$, the current drawn is $I = V / R = 12 \text{ V} / 0.9 \Omega \approx 13.33 \text{ A}$.
This exceeds typical automotive practical limits (e.g., short circuits), so use the given choices. The stated bulb wattage of 40 W is highly implausible for a 12 V system, as a 12 V / 40 W bulb would require $R \approx 0.9 \Omega$, which gives $I \approx 13.33 \text{ A}$; too high.

Therefore, the correct answer, based on the standard calculation and implied unlikelihood of the stated power, is: C. 3.75 amps, 3.2 ohms.

Correct Answer: C.

</think>

<answer>

C: 3.75 amps, 3.2 ohms

</answer>

(d) *Dense*, incorrect

Figure 18: Correct and Incorrect Reasoning Reward for Qwen2.5-7B on MMLU-PRO. Dense reward on correct and incorrect generations using the *interval* and *dense* reasoning reward model.



Figure 19: Correct and Incorrect Reasoning Reward for L1ama3.1-8B on MMLU-PRO. Dense reward on correct and incorrect generations using the *interval* and *dense* reasoning reward model.

D.6.3 MEDREASON

Question:
Shock causes:
Answer Choices:
A. Stagnant hypoxia
B. Anemic hypoxia
C. Hypoxic hypoxia
D. Histotoxic hypoxia

Reasoning + Answer:
mobx
Shock leads to inadequate blood flow to tissues, resulting in insufficient oxygen delivery to cells. This is primarily due to reduced perfusion, not a decrease in oxygen content of the blood or impaired cellular utilization.

Let's evaluate each option:

A. Stagnant hypoxia &g This occurs when blood flow to tissues is reduced (e.g., due to poor perfusion), leading to oxygen deprivation despite adequate oxygen content in the blood. This is exactly what happens in shock, where circulation is impaired and blood flow is stagnant. This is the most appropriate answer.

B. Anemic hypoxia &g This results from a decrease in hemoglobin or red blood cells (e.g., hemorrhage, hemolytic anemia), leading to reduced oxygen-carrying capacity. While hemorrhagic shock may involve anemia, the primary mechanism of hypoxia in shock is not anemia per se, but rather poor perfusion.

C. Hypoxic hypoxia &g This occurs when oxygen content in the blood is low due to low partial pressure of oxygen (e.g., in pulmonary disorders). This is not the primary mechanism in shock.

D. Histotoxic hypoxia &g This occurs when cells cannot utilize oxygen (e.g., cyanide poisoning). It is not related to shock.

In shock, the main problem is that tissues receive insufficient blood flow, leading to stagnant hypoxia. The blood may have normal oxygen content, but it's not reaching the tissues due to poor circulation.

<answer>
A: Stagnant hypoxia
</answer>

(a) Interval, correct

Question:
Shock causes:
Answer Choices:
A. Stagnant hypoxia
B. Anemic hypoxia
C. Hypoxic hypoxia
D. Histotoxic hypoxia

Reasoning + Answer:
mobx
Shock leads to inadequate tissue perfusion, resulting in reduced oxygen delivery to tissues. The primary issue is that oxygen cannot reach tissues adequately due to poor blood flow, even if oxygen content in the blood is normal. This condition is classified as hypoxic hypoxia, where the tissue is deprived of oxygen because of insufficient oxygen delivery due to impaired perfusion.

Let's evaluate the options:

A. Stagnant hypoxia &g This occurs when blood flow is reduced but oxygen content is normal, and the tissue is hypoxic due to poor blood flow. It is a form of hypoxic hypoxia and can be seen in shock, but the term "stagnant hypoxia" is more specific to reduced blood flow without oxygen content change. However, in shock, the main issue is reduced delivery, which is more precisely described as hypoxic hypoxia.

B. Anemic hypoxia &g This occurs due to reduced hemoglobin (e.g., hemorrhage), leading to decreased oxygen-carrying capacity. While hemorrhagic shock can cause anemia, the core pathophysiology of shock is not primarily anemia per se.

C. Hypoxic hypoxia &g This is defined by reduced oxygen tension in the tissues due to inadequate oxygen delivery from the blood, typically due to poor perfusion. This is the hallmark of shock. The tissues are hypoxic because blood flow is insufficient, even if oxygen content is normal.

D. Histotoxic hypoxia &g This occurs when cells cannot utilize oxygen (e.g., cyanide poisoning), not due to delivery issues.

In shock, the primary mechanism is reduced perfusion leading to inadequate oxygen delivery to tissues, which is the definition of hypoxic hypoxia.

<answer>
C: Hypoxic hypoxia
</answer>

(b) Interval, incorrect

Figure 20: Correct and Incorrect Reasoning Reward for Qwen3-4B on MEDREASON. Dense reward on correct and incorrect generations using the *dense* reasoning reward model.

Question:
Shock causes:
Answer Choices:
A. Stagnant hypoxia
B. Anemic hypoxia
C. Hypoxic hypoxia
D. Histotoxic hypoxia

Reasoning + Answer:
<think>
To determine the correct answer, let's analyze the options given. Shock can be caused by various factors, including inadequate blood flow, leading to insufficient oxygen delivery to tissues. Stagnant hypoxia (A) occurs when there is a lack of blood flow due to cardiac dysfunction, resulting in inadequate oxygen delivery to tissues. This is a common cause of shock, particularly in conditions such as cardiogenic shock.

Anemic hypoxia (B) is caused by a decrease in the amount of hemoglobin or red blood cells, which reduces the capacity of the blood to carry oxygen. While this can lead to tissue hypoxia, it is not the primary cause of shock.

Hypoxic hypoxia (C) occurs when there is a decrease in the partial pressure of oxygen in the air or in the blood. This can be caused by respiratory or altitude-related issues but is not directly related to the causes of shock.

Histotoxic hypoxia (D) is caused by the presence of substances that interfere with the body's ability to use oxygen, such as certain toxins. While this can lead to tissue hypoxia, it is not a direct cause of shock.

Considering the options, stagnant hypoxia is a direct cause of shock due to its impact on blood flow and oxygen delivery to tissues.

</think>
<answer>
A: Stagnant hypoxia
</answer>

(a) interval, correct

Question:
Shock causes:
Answer Choices:
A. Stagnant hypoxia
B. Anemic hypoxia
C. Hypoxic hypoxia
D. Histotoxic hypoxia

Reasoning + Answer:
<think>
To answer this question, we need to identify the cause of shock. Shock is a systemic condition characterized by inadequate perfusion of vital organs, leading to organ dysfunction. It can result from various causes, including inadequate oxygen delivery to the tissues (hypoxia).

A. Stagnant hypoxia refers to a situation where there is impaired blood flow, leading to inadequate oxygen delivery to tissues. This can be caused by cardiac arrest, severe cardiac dysfunction, or significant blood loss. However, this is not the primary cause of shock.

B. Anemic hypoxia occurs when the blood lacks sufficient red blood cells or hemoglobin, impairing oxygen delivery to tissues. This can be caused by an anemia, severe bleeding, or carbon monoxide poisoning. While anemia can contribute to shock, it is not the most direct cause.

C. Hypoxic hypoxia refers to inadequate oxygen levels in the air breathed in (hypoxia) or a failure of the lungs to exchange oxygen properly. This can be caused by high altitudes, respiratory failure, or carbon monoxide poisoning. Hypoxic hypoxia is a direct cause of shock, as it impairs the body's ability to deliver oxygen to tissues.

D. Histotoxic hypoxia occurs when certain toxins or substances interfere with the body's ability to use oxygen. This can be caused by carbon monoxide poisoning, cyanide poisoning, or other toxins. While histotoxic hypoxia can contribute to shock, it is not the primary cause.

Considering the options, the most direct cause of shock is related to inadequate oxygen delivery to tissues, which is best described by option C. Hypoxic hypoxia is a direct cause of shock, as it impairs the body's ability to deliver oxygen to tissues.

</think>
<answer>
C: Hypoxic hypoxia
</answer>

(b) interval, incorrect

Figure 21: Correct and Incorrect Reasoning Reward for Llama3.1-8B on MEDREASON. Dense reward on correct and incorrect generations using the *interval* reasoning reward model.